

Generative AI for art, images, and video

AMCC5160

Lecture 04

22 September 2026



Five readings, one creative process

- 01** Video models
- 02** Visual art
- 03** Filmmaking
- 04** Visual computing
- 05** Creative industries

What changes when an image moves?



"The ball is in blue fire and turns into a dragon in fire."



"A man rides on a huge fish, flying from the water into the sky."

A still image leaves time open.
A video chooses what happens next.

Our running example

An empty studio.

A painting begins to move.

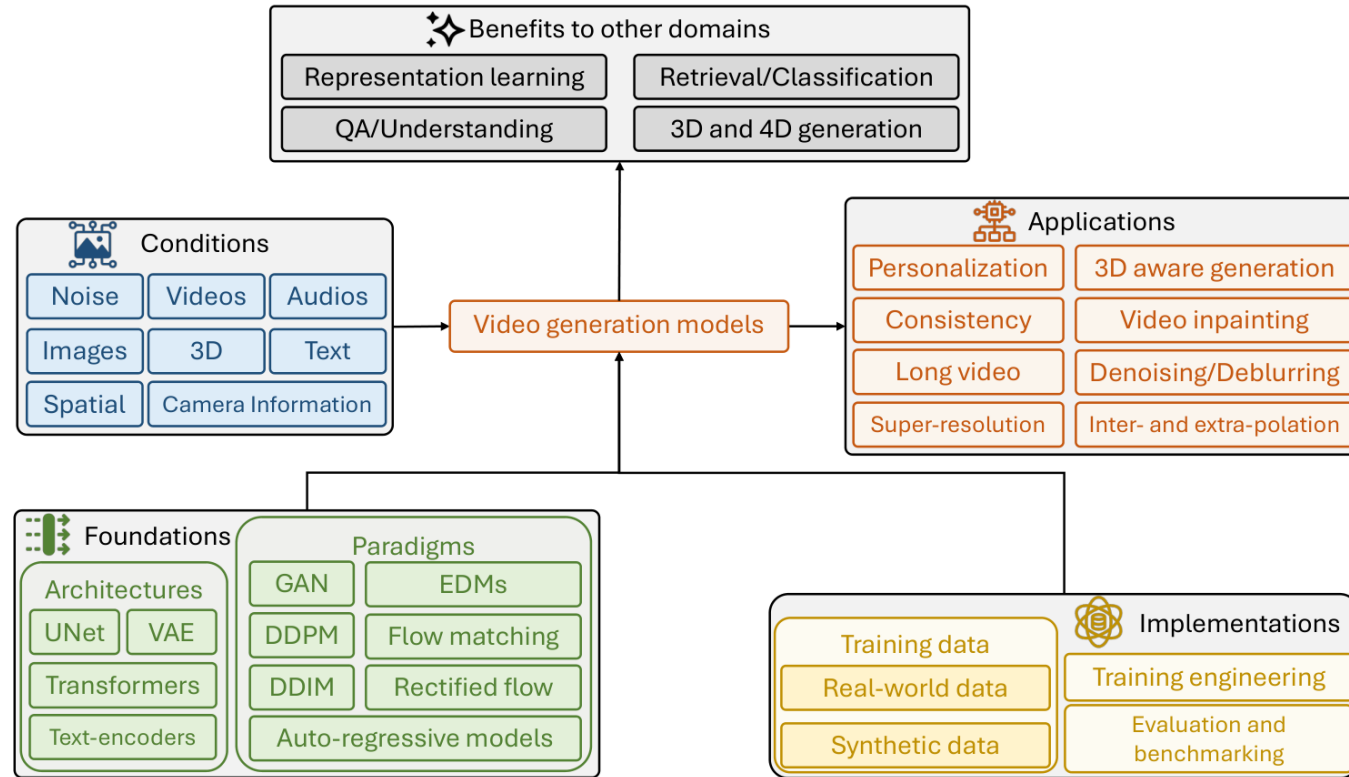
Someone returns.

01

Video generation models

Wang et al.
Survey of Video Diffusion Models
2025, revised February 2026

The field at a glance



Inputs guide a model.
The model produces a sequence.

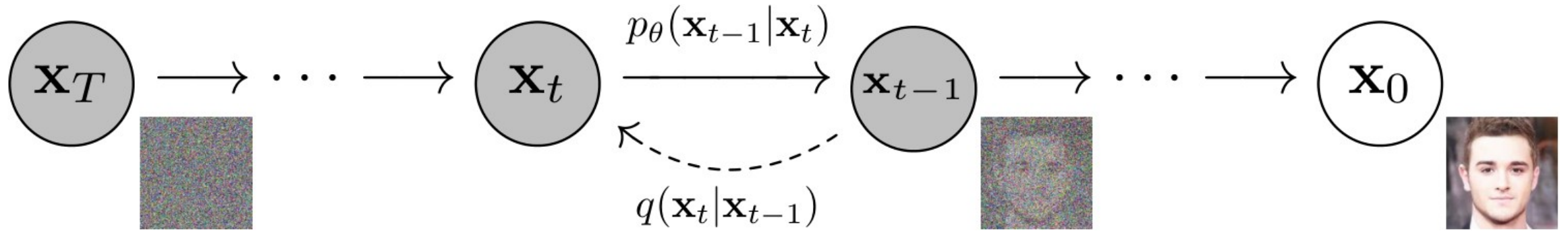
Three routes to a video

Text describes a scene.

An image fixes a starting appearance.

A video supplies existing movement.

Diffusion in one picture



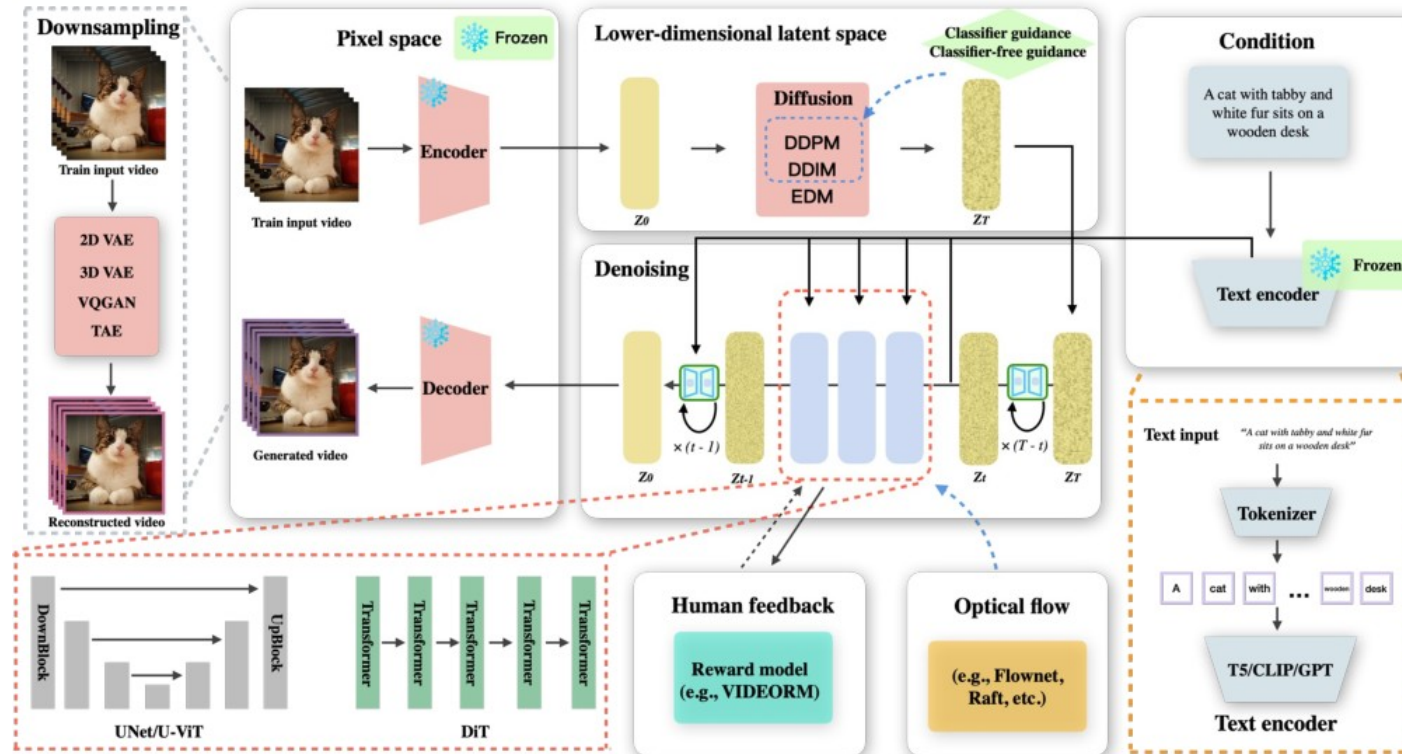
Training teaches noise removal.
Generation refines a noisy sample.

A video has two kinds of time

Frame time: when an event happens.

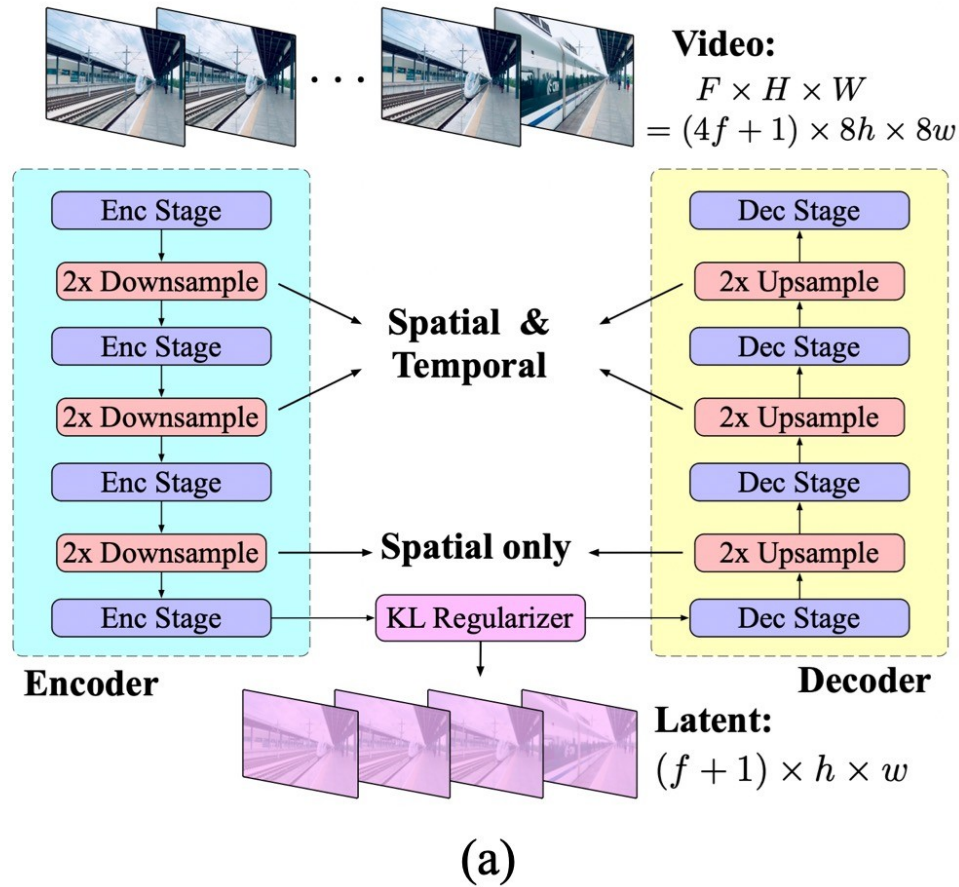
Denoising time: how a sample improves.

The generation pipeline



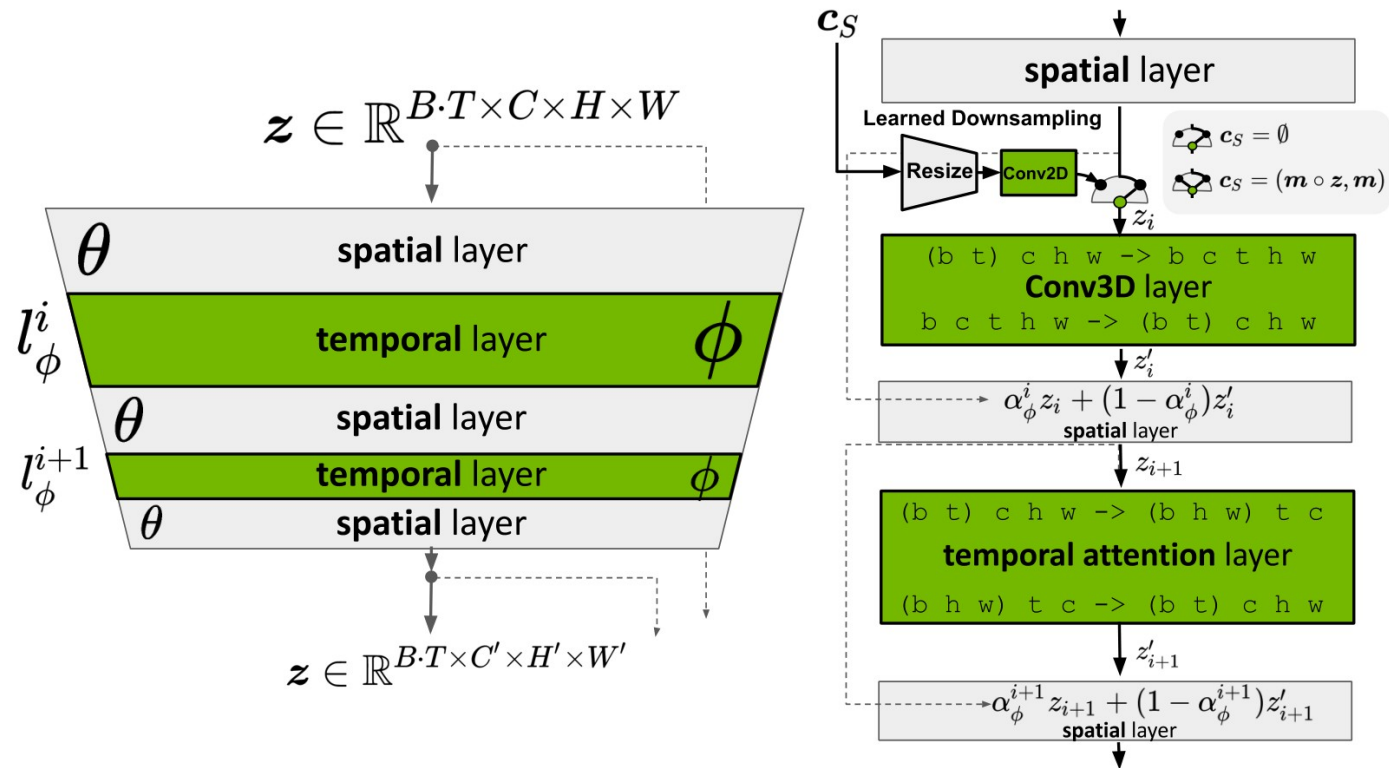
A prompt guides refinement in a compressed space.

A latent is a compressed representation



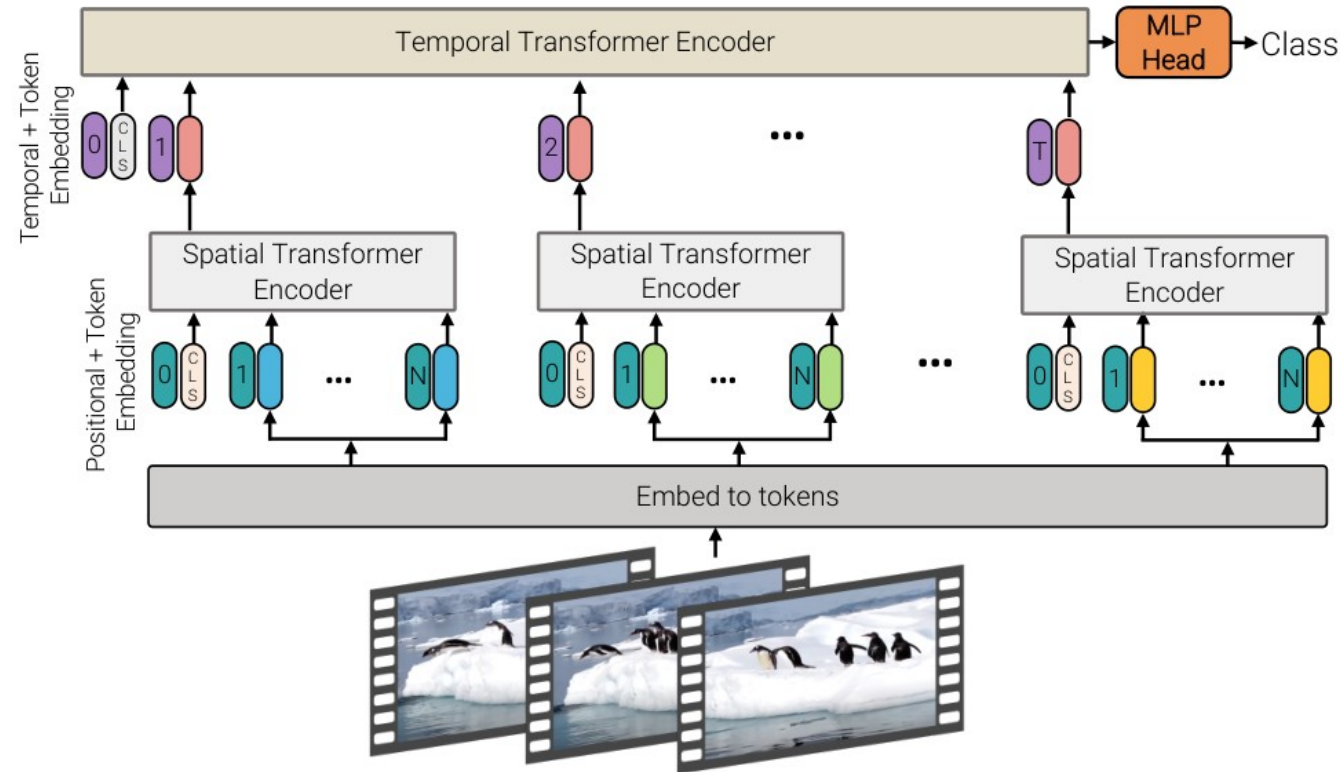
The model works with fewer values than raw pixels.

Spatial attention and temporal attention



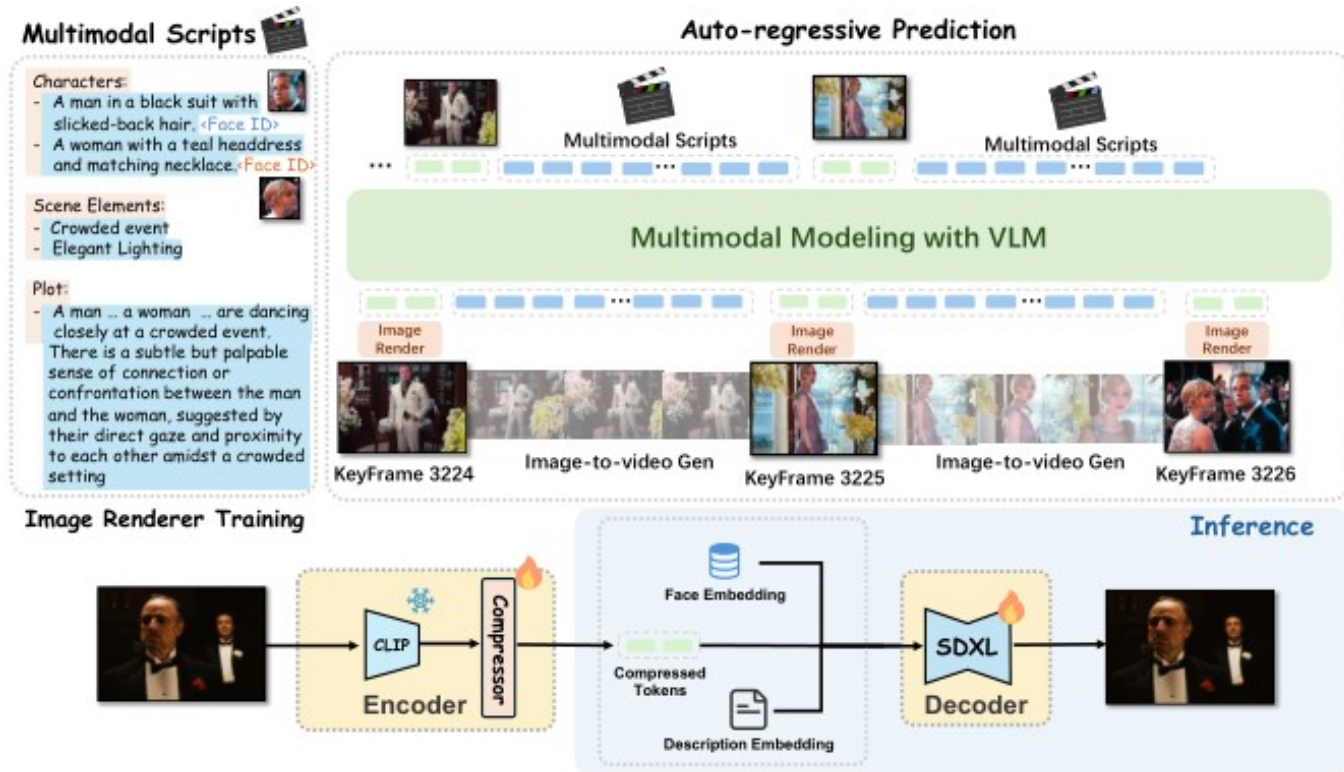
Spatial: relationships within a frame.
Temporal: relationships across frames.

A second way to connect frames



A frame encoder handles appearance.
A temporal encoder connects the sequence.

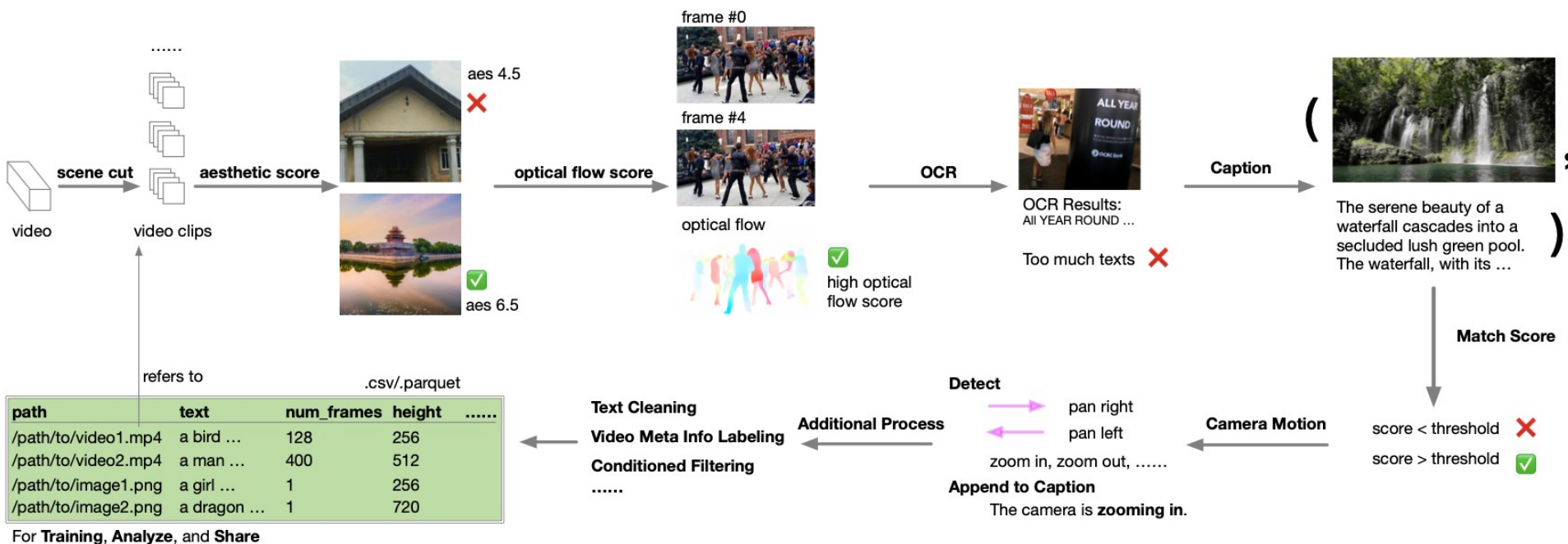
Predicting a sequence in parts



Autoregressive methods build on earlier outputs.

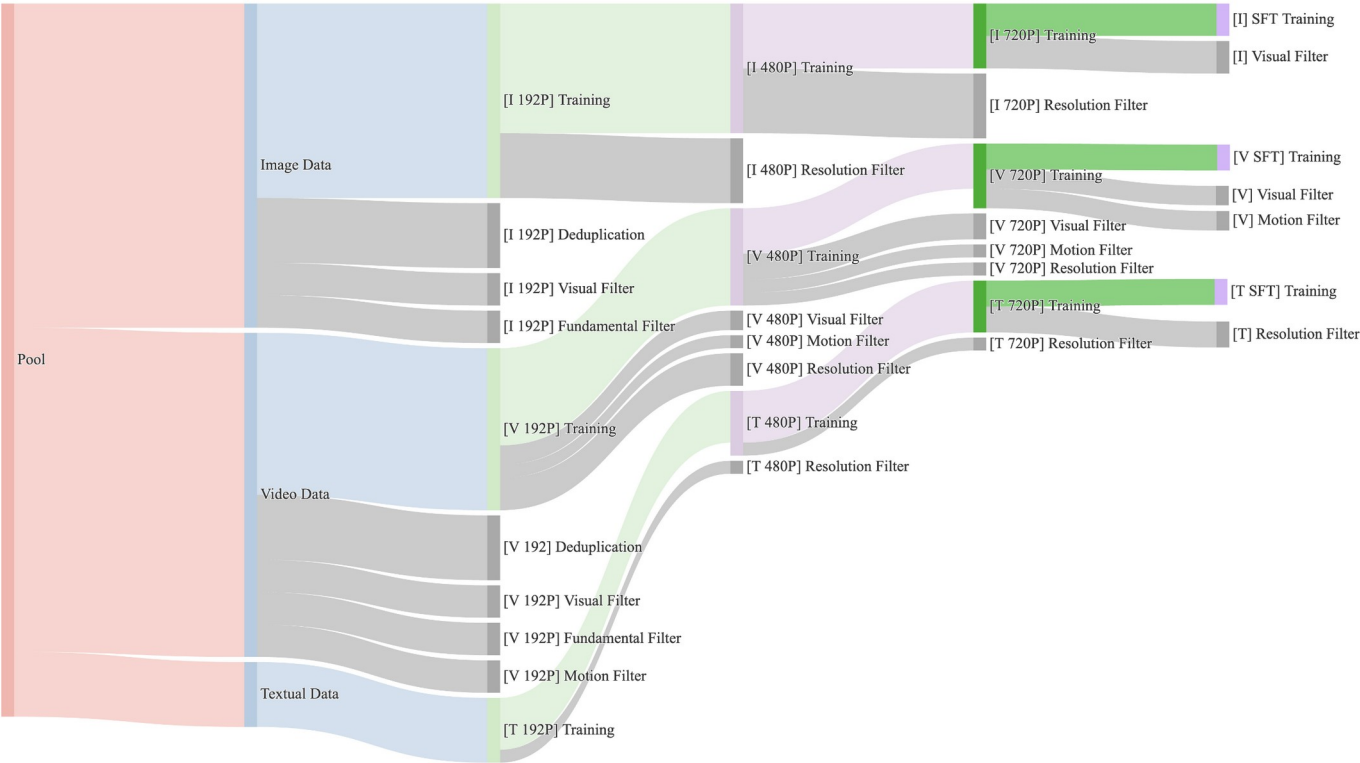
Training data shapes the result

Open-Sora Data Processing Pipeline v1.1



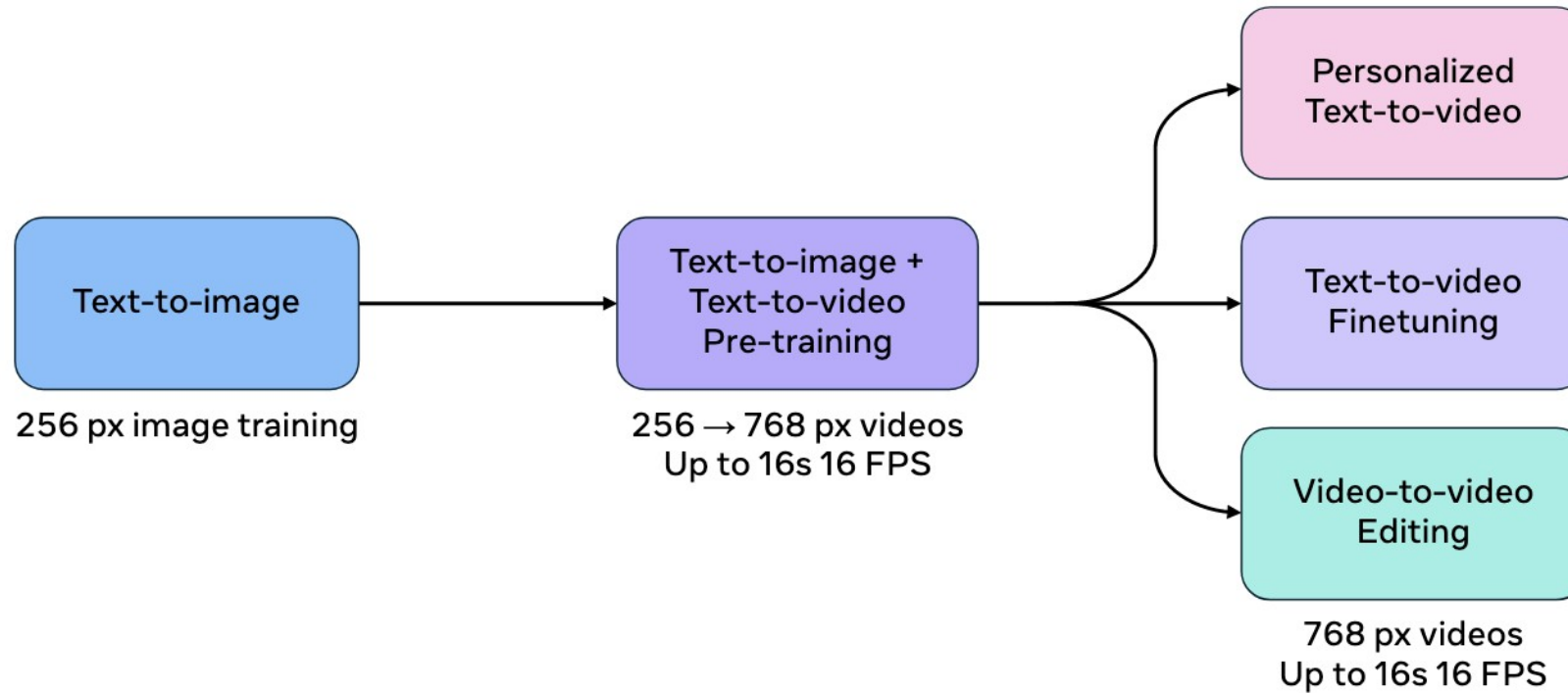
Selection and captions affect what a model can learn.

A closer look at data filtering



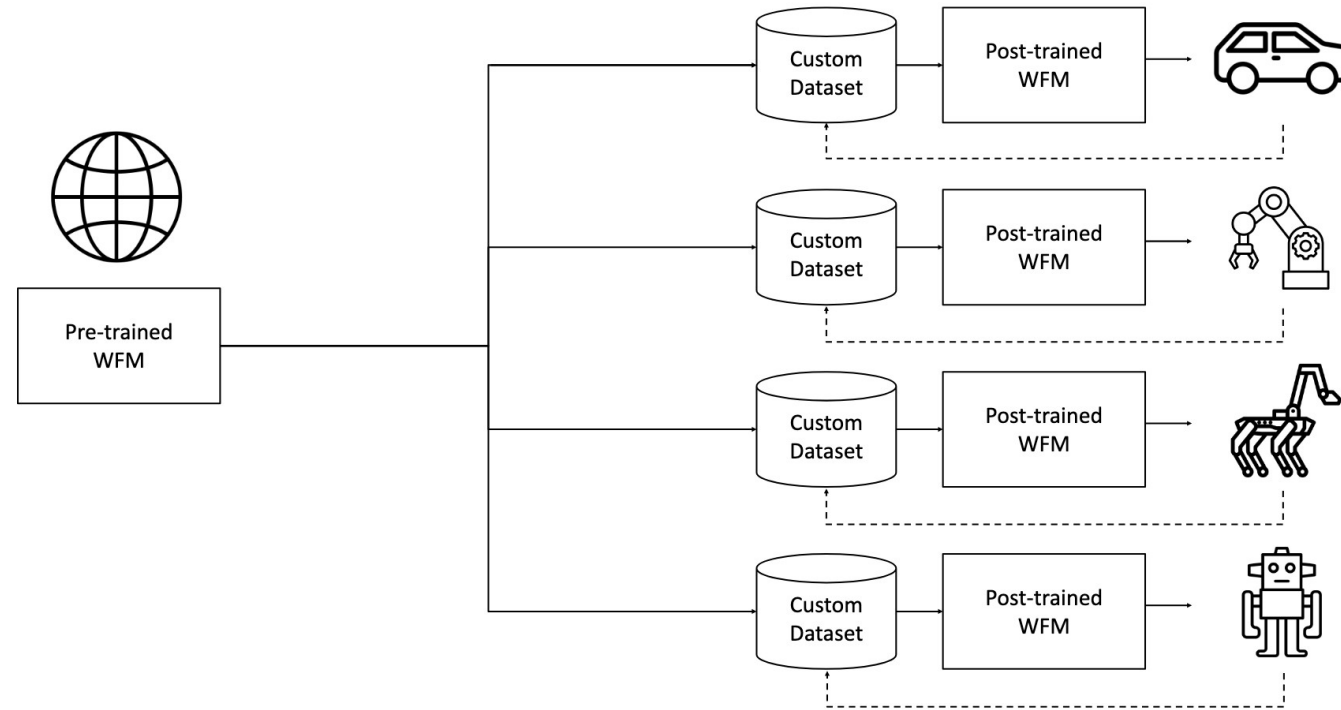
Filtering trades coverage against usable training examples.

One foundation, several tasks



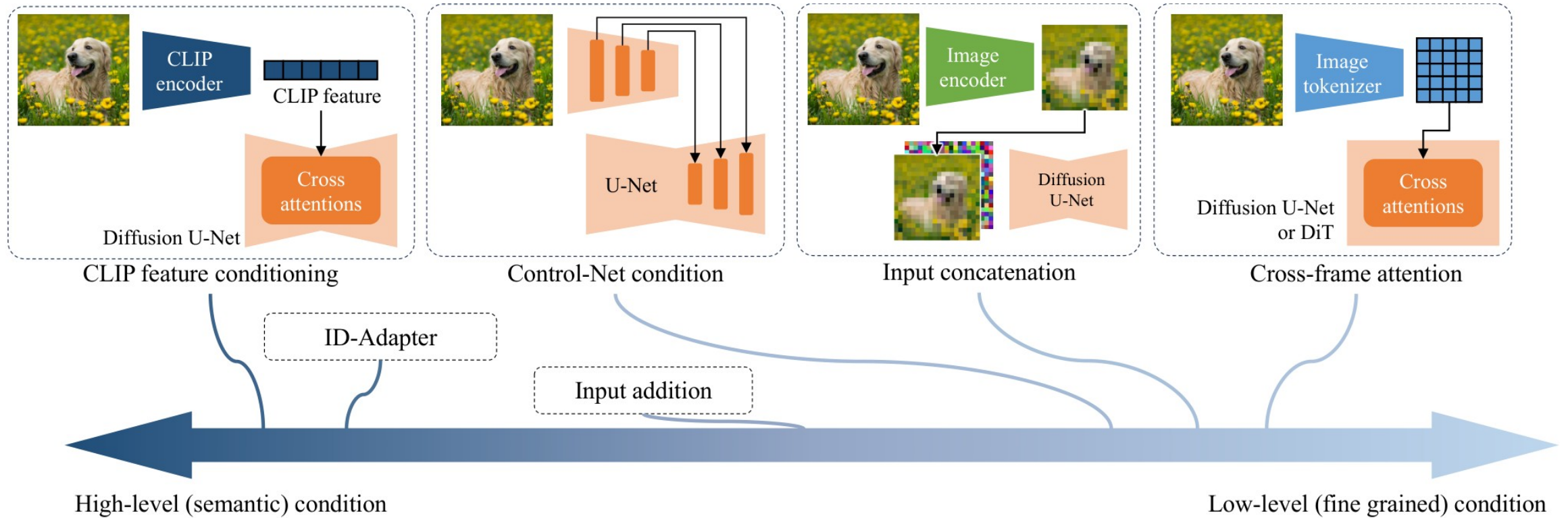
Generation, personalization, and editing can share a base.

A foundation model can adapt to a domain



Further training specializes a general model.

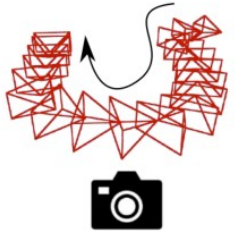
References carry different information



A reference can suggest meaning, layout, or appearance.

Motion and camera control

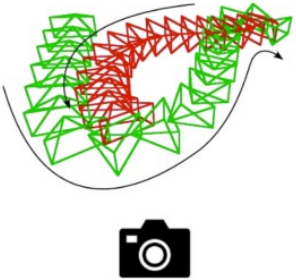
Pose Sequence



frame



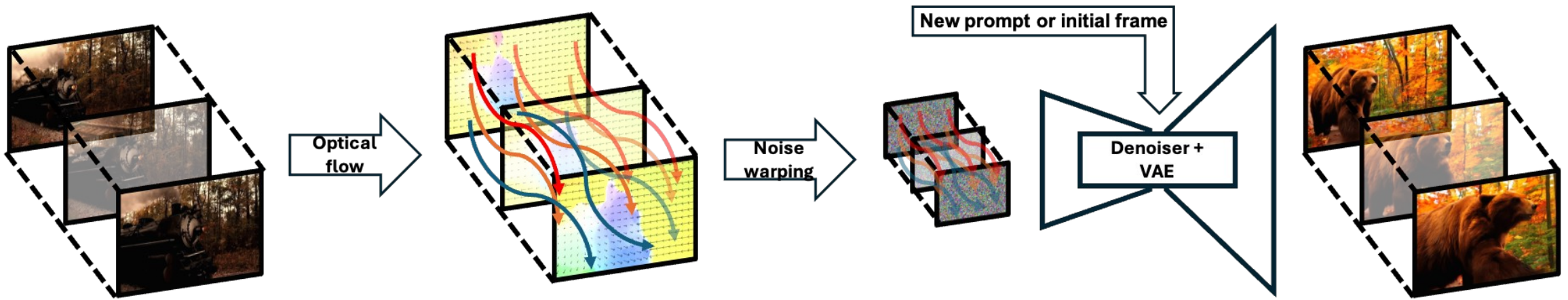
Entity **a flamingo with pink feathers, long slender legs, a curved neck, and a tuft of feathers on its back** Location **city**



Entity **a gazelle with a body covered in sleek tan fur, long legs, and elegant curved horns** Location **glacier**
a man with medium-length straight brown hair, tall and slender, a gray crew-neck t-shirt, beige trousers

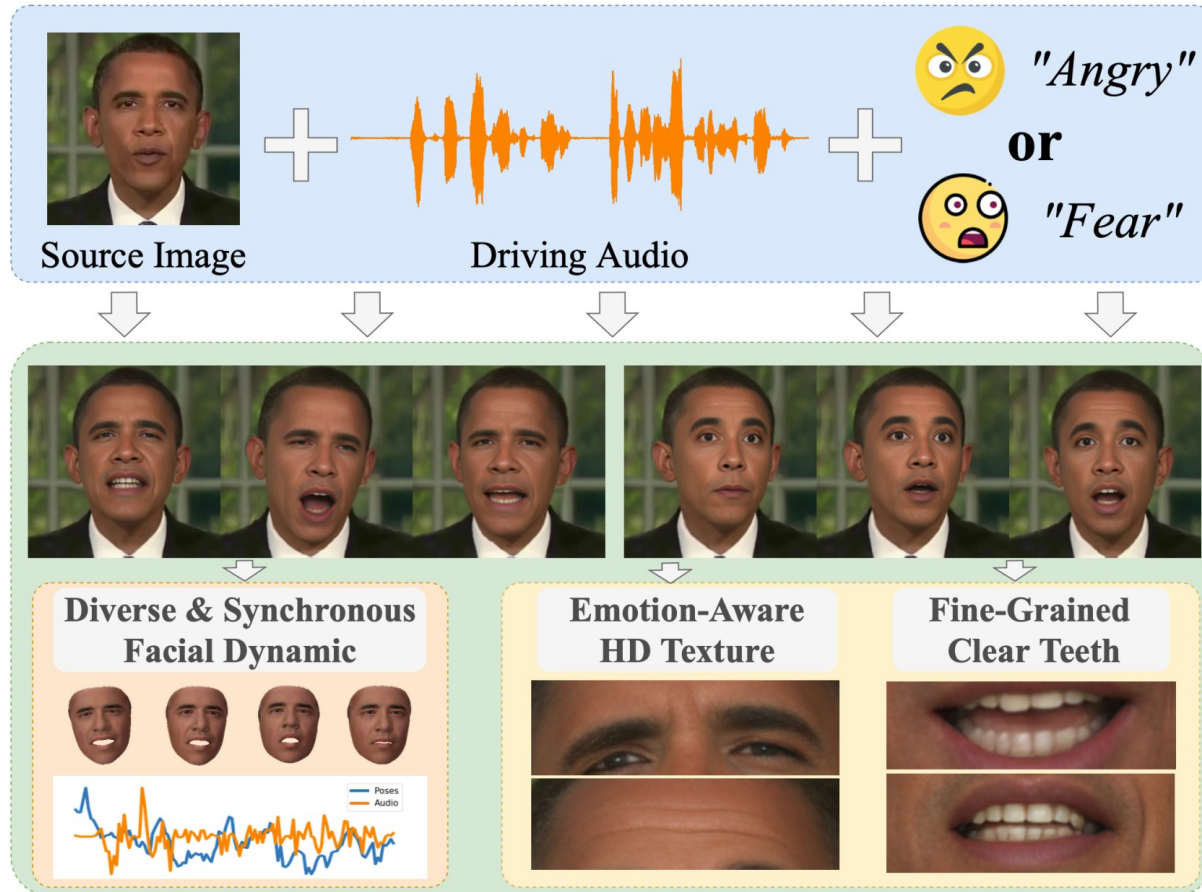
An actor can move while the camera stays still.
A camera can move around a still actor.

Motion as a field



Optical flow describes how image locations move.

Sound can guide a performance



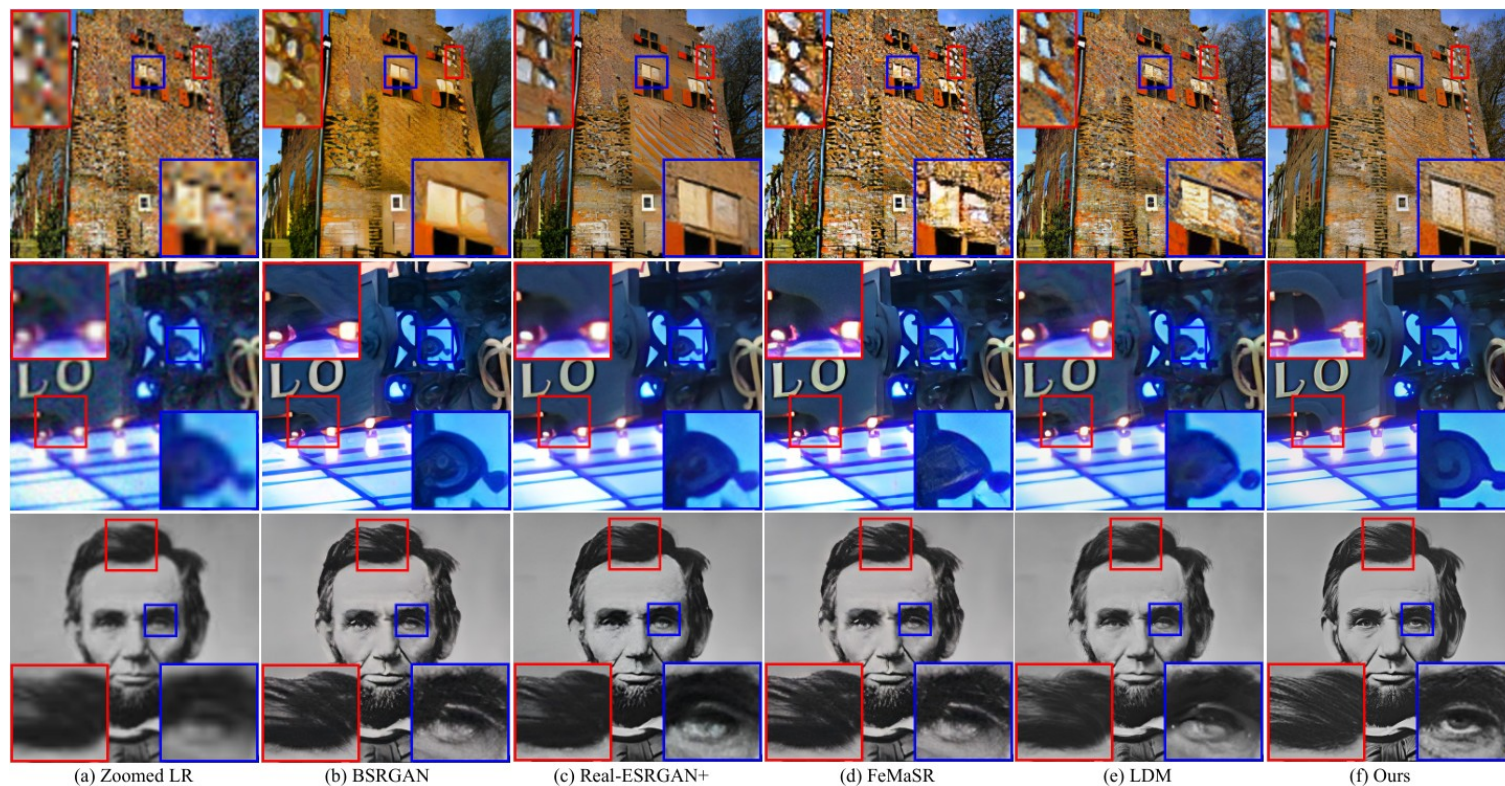
Audio can constrain mouth movement and timing.

Editing a region through time



The selected change must persist across frames.

Sharper detail can change what we see



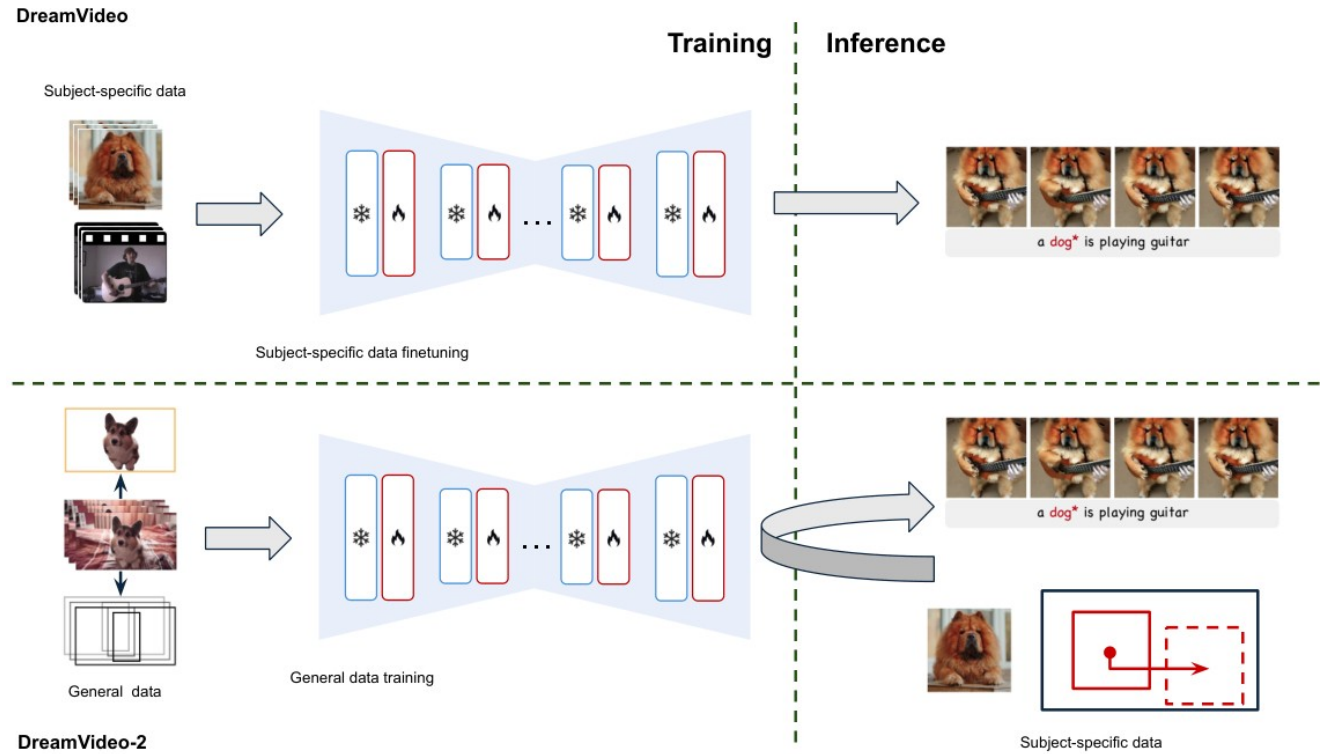
Plausible texture may differ from the original scene.

Before, between, and after



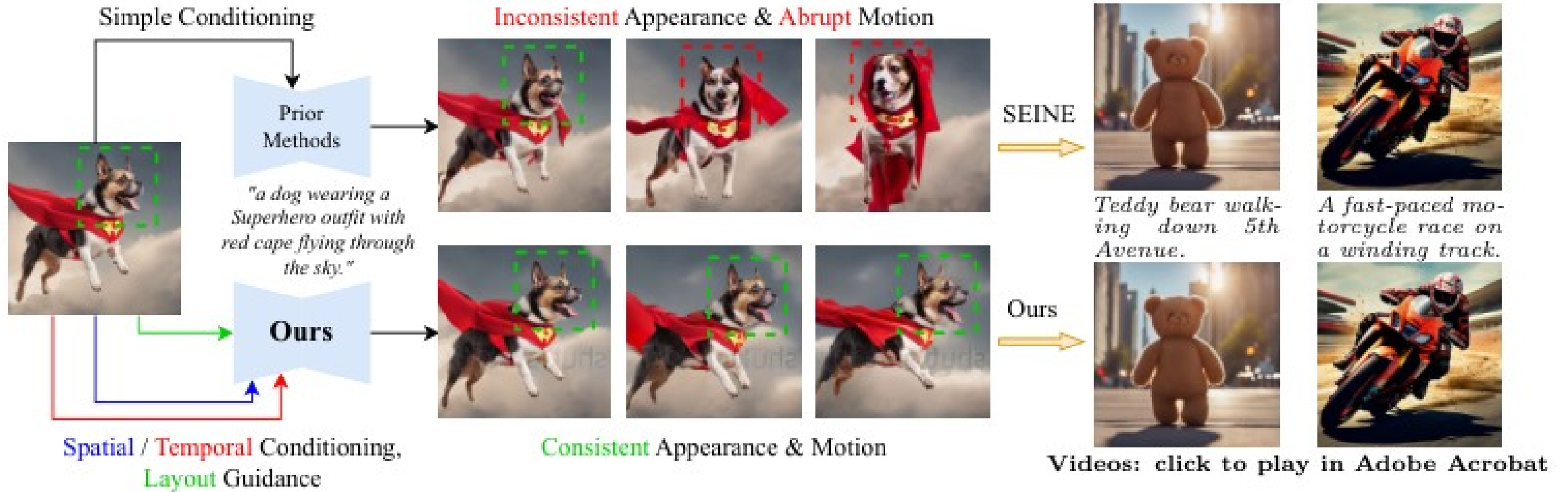
Prediction extends a sequence.
Interpolation fills a gap.

A subject that stays recognizable



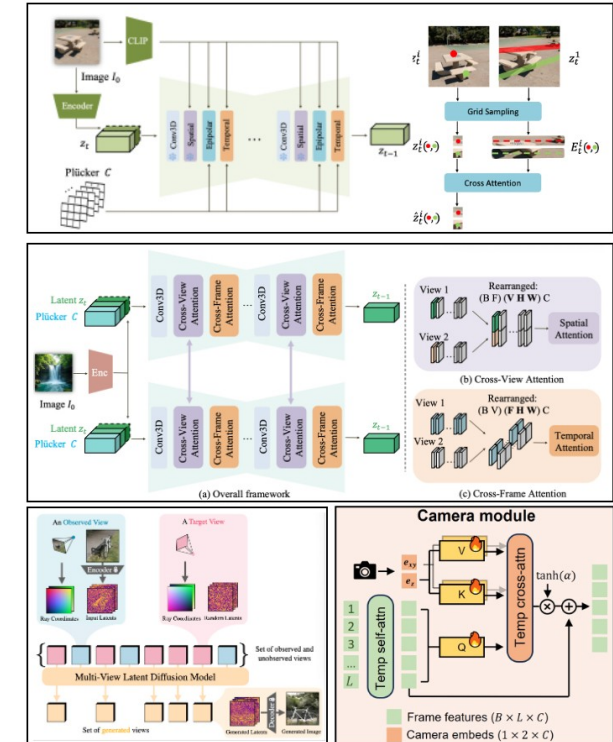
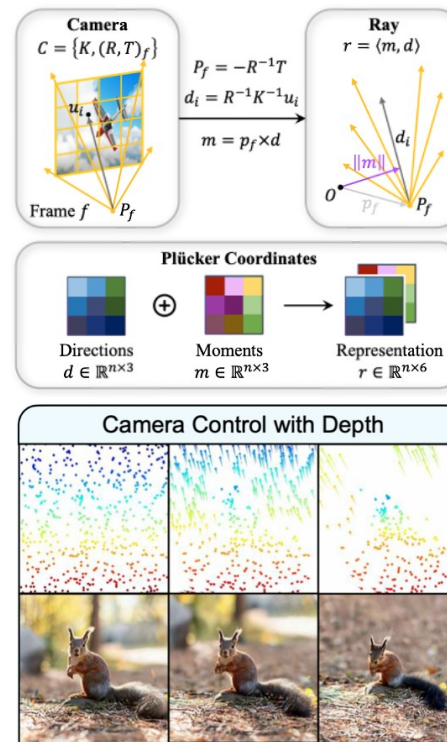
Identity needs to survive changes in pose and view.

Continuity needs more than a good first frame



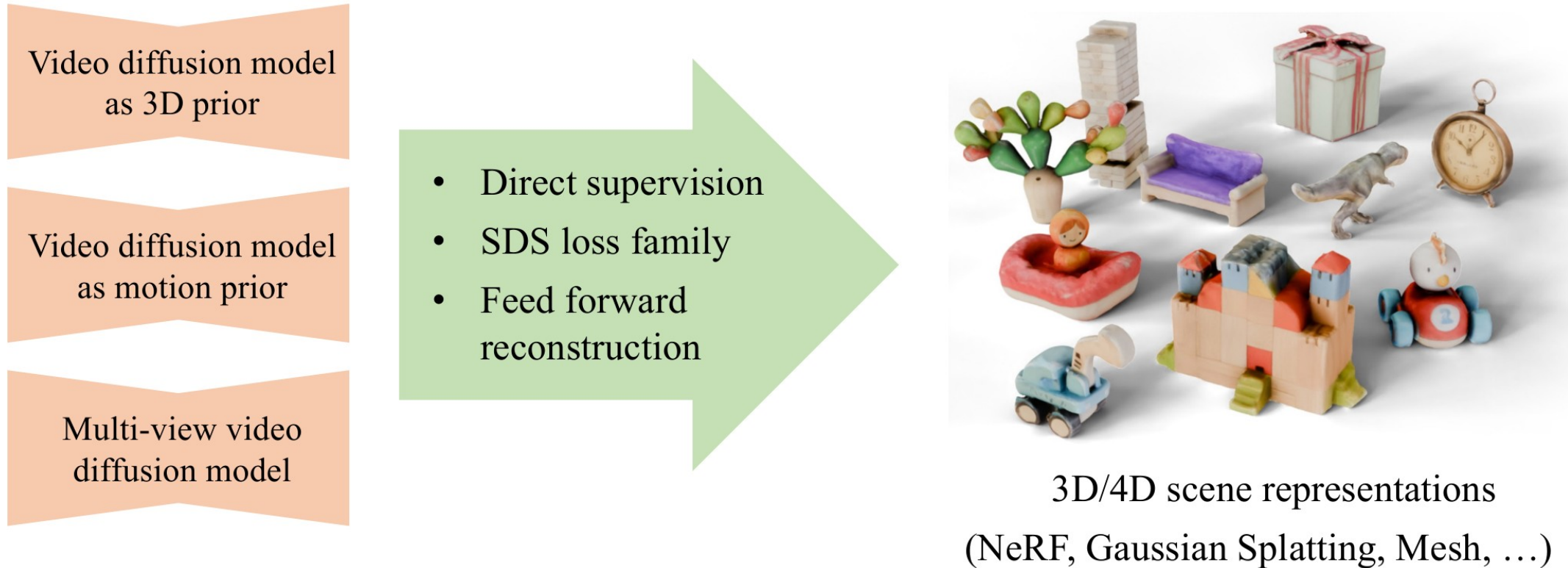
Check appearance and movement across the whole shot.

Video models can incorporate 3D information



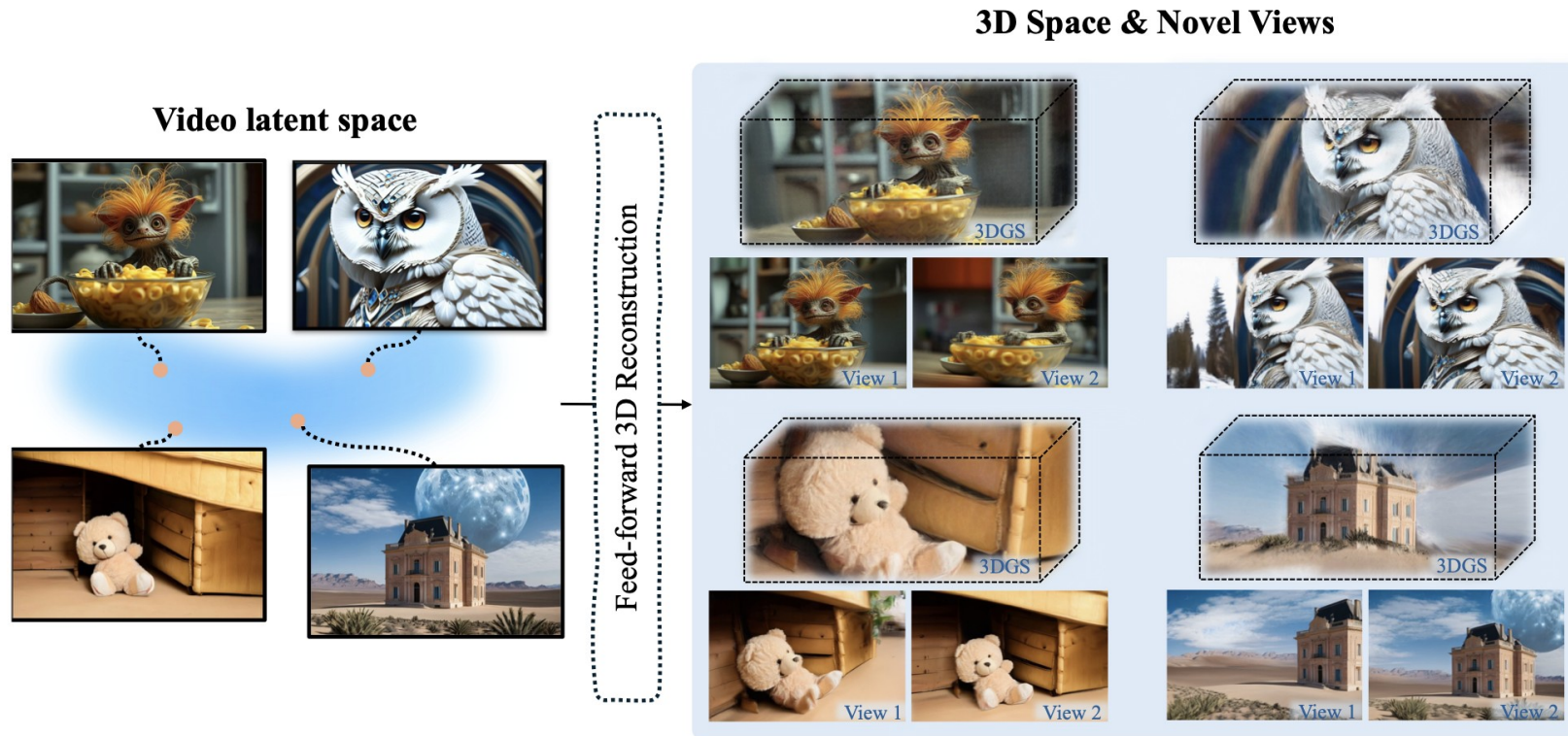
Camera information links an image to a viewpoint.

Video knowledge can guide a scene representation



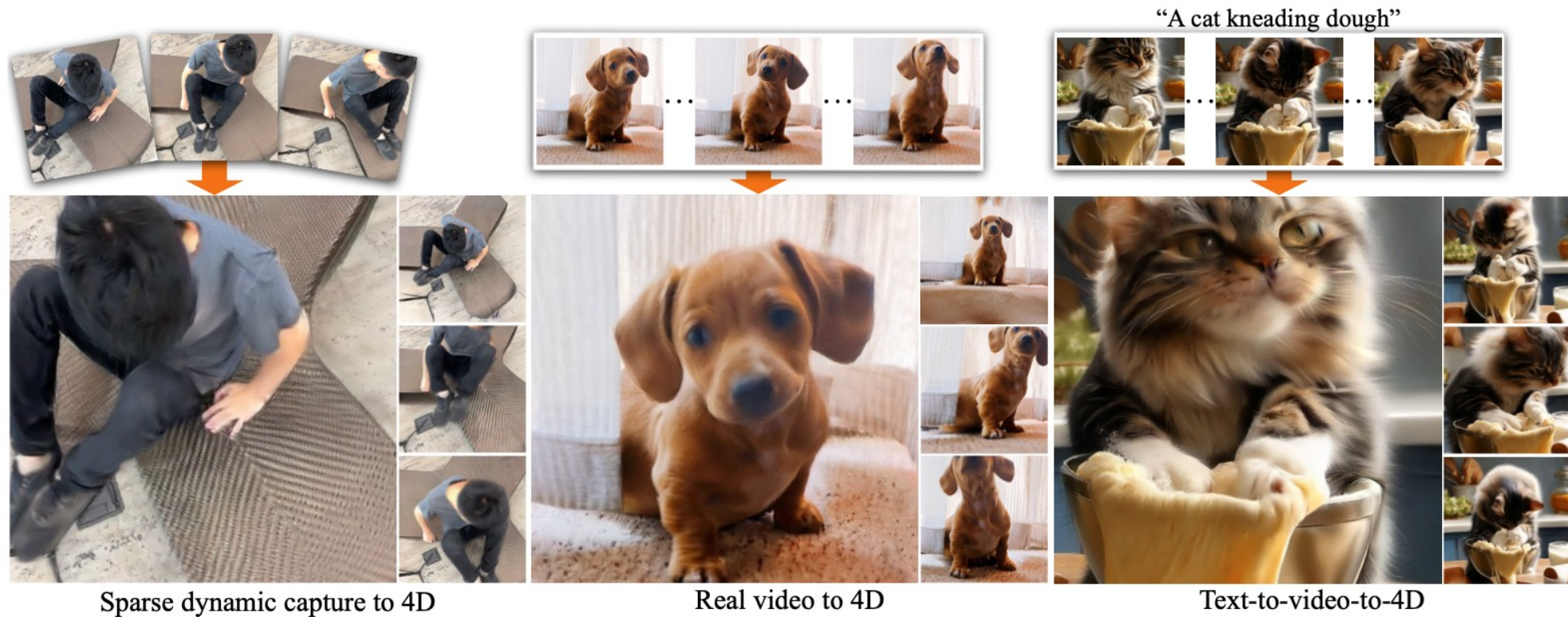
A generated sequence can help build a 3D scene.

Wonderland connects video and reconstruction



A reconstruction model predicts a scene with new views.

A moving scene varies in viewpoint and time



CAT4D models both dimensions.

Video-model checkpoint

What fixes appearance?

What controls movement?

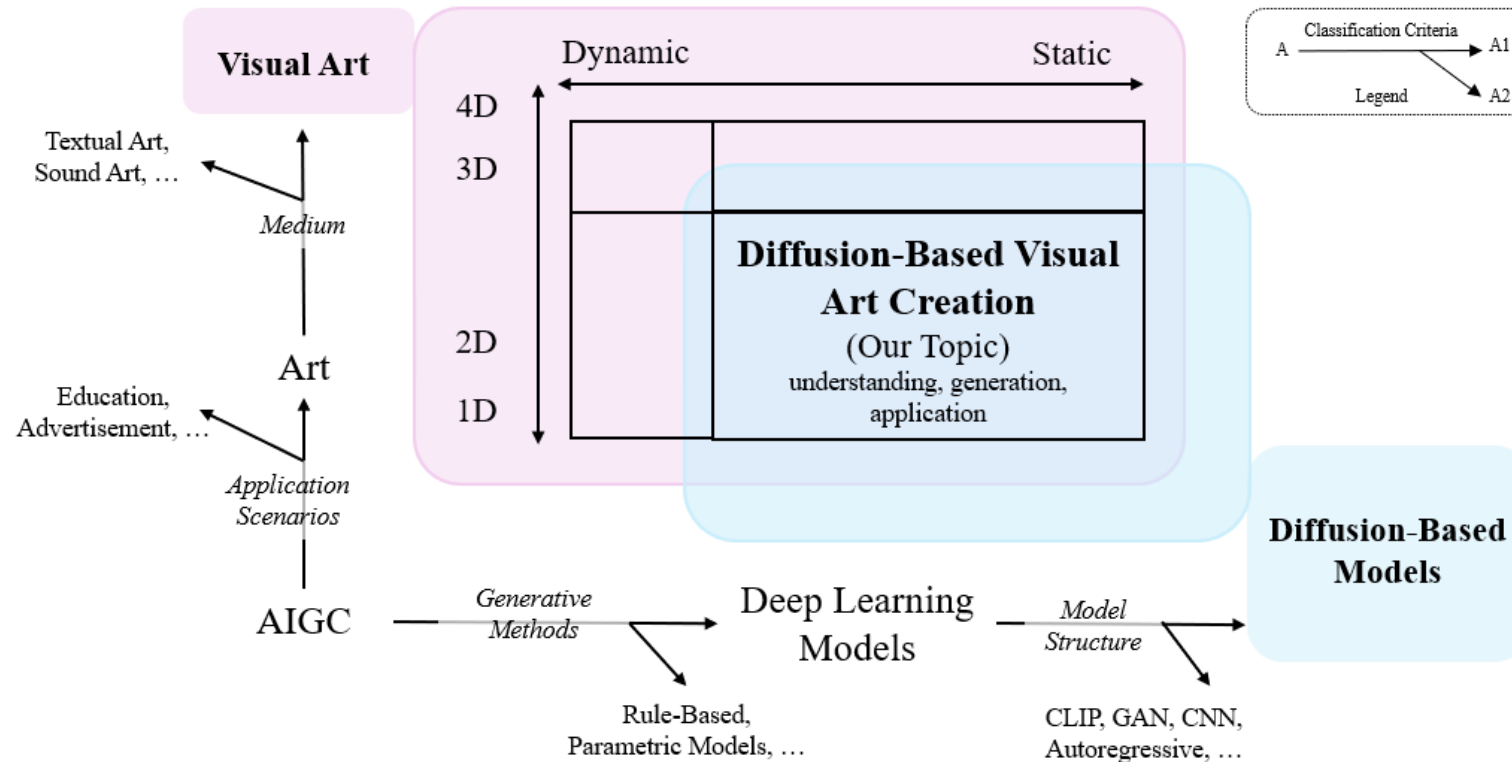
What still needs inspection?

02

Visual art creation

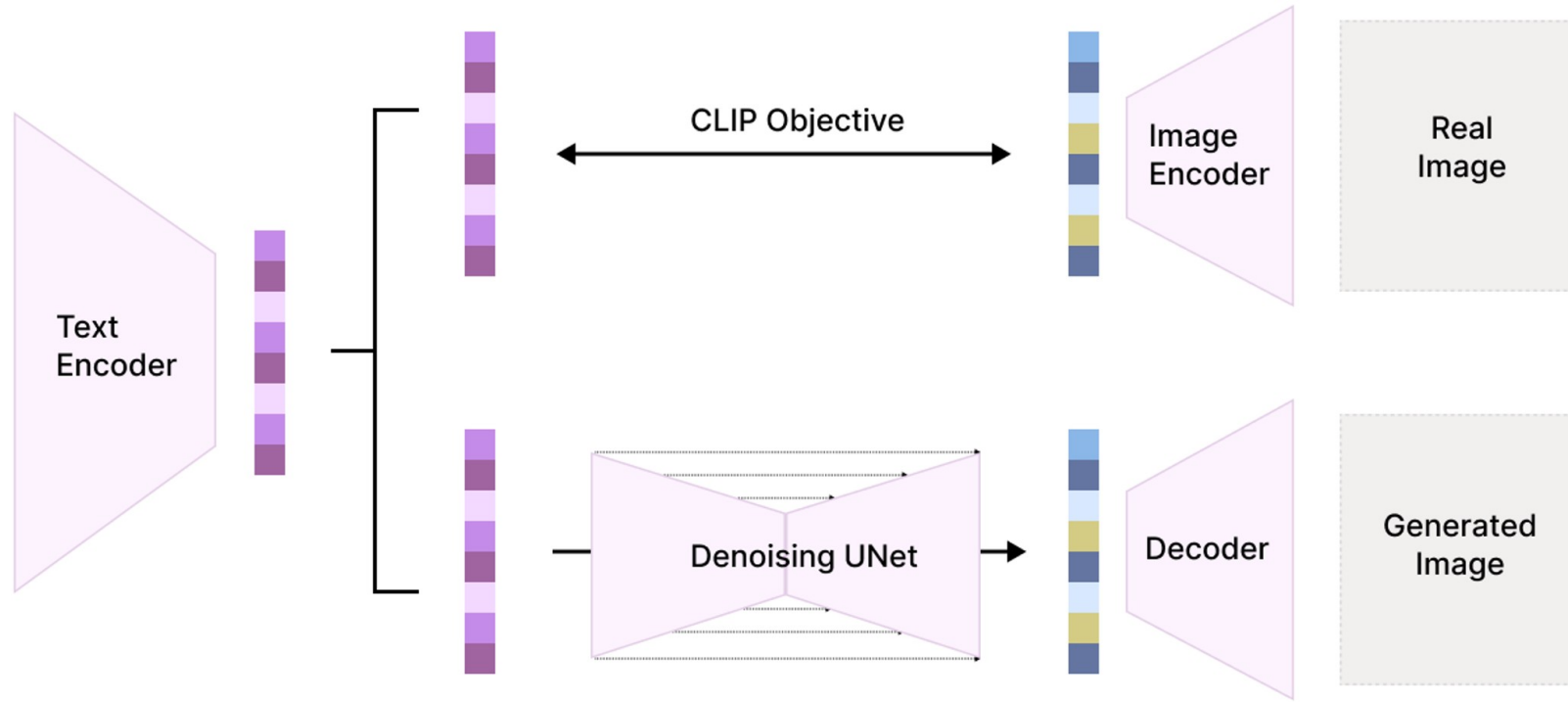
Wang, Chen, and Wang
Diffusion-Based Visual Art Creation
ACM Computing Surveys, 2025

Artistic goals and technical methods



The survey studies where these two perspectives meet.

Text becomes a condition for generation



A text encoder turns words into a model representation.

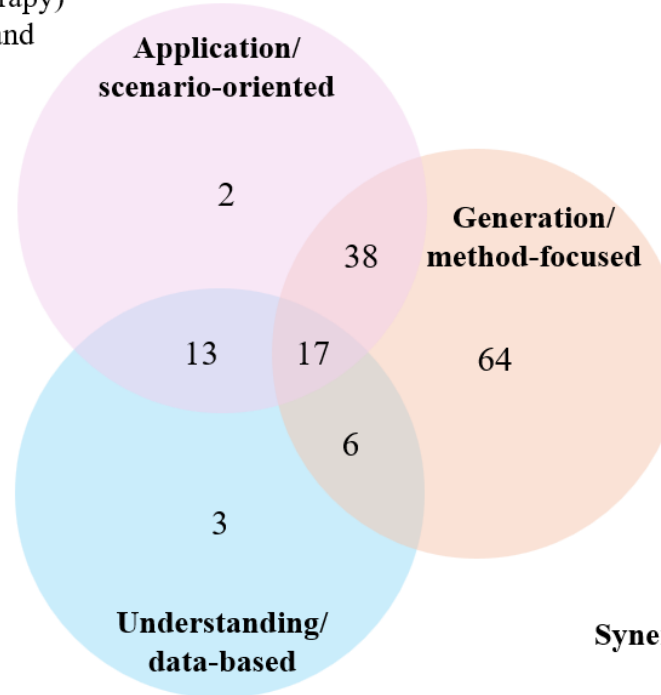
Three research perspectives

- Different application scenarios (e.g., visualization, storytelling, therapy)
- Different art forms/genres and corresponding features



Psychological perspective

- Different datasets, data source and features
- Different data modalities and formats



Technological perspective



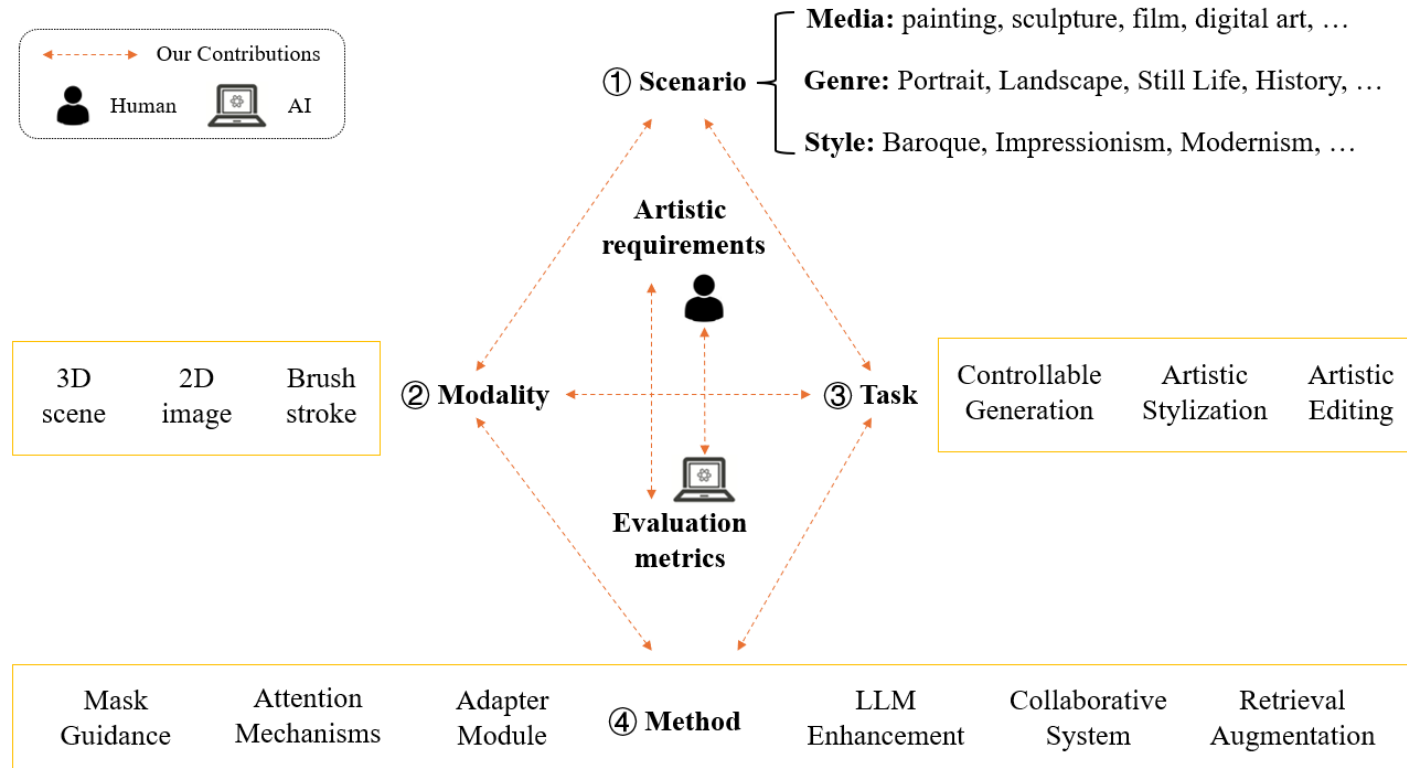
- Different generative tasks (e.g., layout control, style transfer, image editing, ...)
- Different generative methods (e.g., Mask Guidance, Attention Mechanisms, Adapter Module, ...)

Synergistic perspective



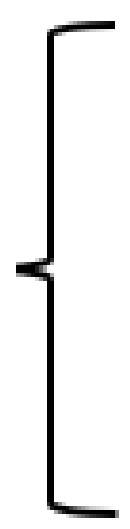
Creating images is one part of the picture.

The artist's intention comes first



A feeling becomes a set of visible decisions.

A scenario gives the work a purpose

- 
- Media:** painting, sculpture, film, digital art, ...
 - Genre:** Portrait, Landscape, Still Life, History, ...
 - Style:** Baroque, Impressionism, Modernism, ...

A portrait, an installation, and a film ask different things.

A modality is the form of the material

3D
scene

2D
image

Brush
stroke

Pixels, brush strokes, and 3D scenes support different edits.

Four useful creative tasks

Generate content

Control a result

Change a style

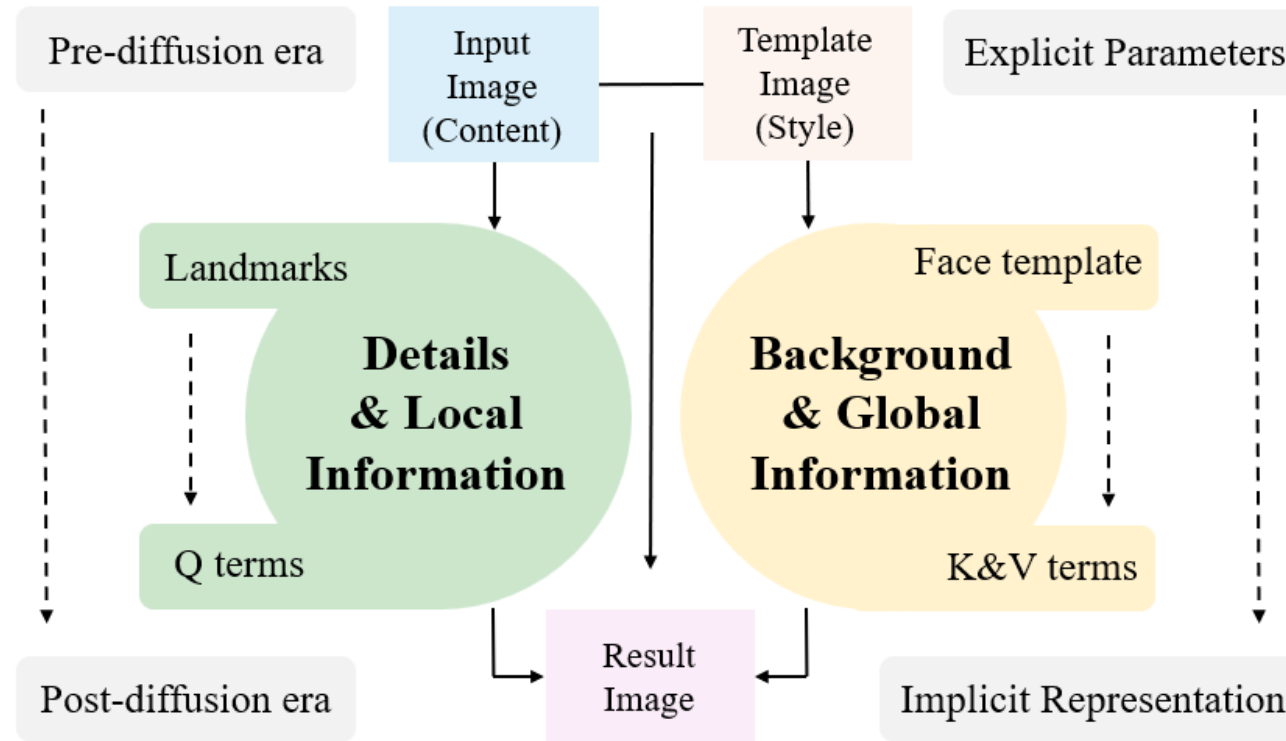
Edit selected content

An edit needs a boundary

A mask specifies where a change may occur.

The surrounding image provides context.

Style includes structure



Shape and composition matter alongside color.

Content and style need separate decisions

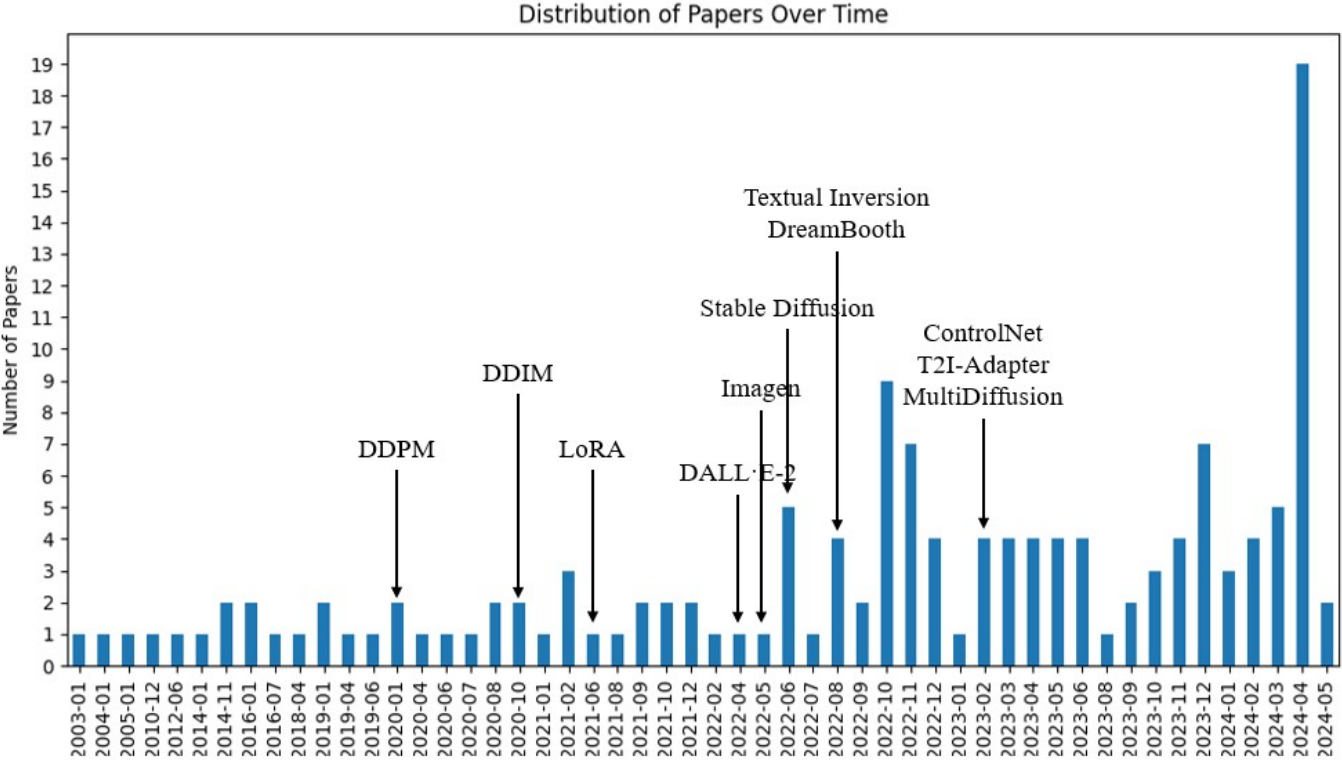
Keep the subject recognizable.

Choose which aspects of appearance may change.

A technical score answers a narrow question

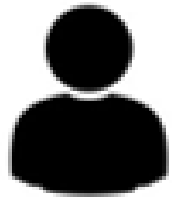
Matching a prompt does not establish artistic success.

The literature changed after diffusion



The survey records growth in its selected research corpus.

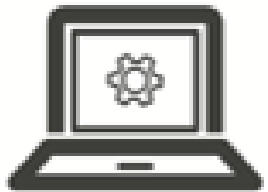
Human involvement changes across a workflow



Creator → Designer → Optimizer --> Consumer

Human Design
AI assistance

Human-AI
Collaboration



Analyzer → Assistant → Generator --> Creator

Choosing, revising, and judging remain creative acts.

Our studio scene as an artwork

Intention: an uneasy memory.

Decision: the painting changes before the room does.

One intention, one intervention

Choose a visual decision.

Explain how an audience might read it.

Art becomes a sequence

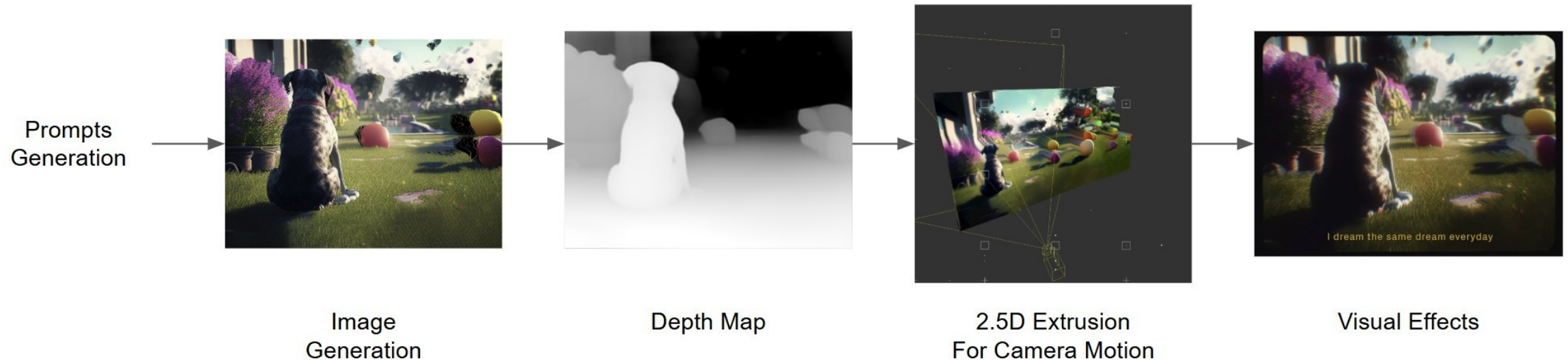
The next question is how the choices survive across shots.

03

Film creation

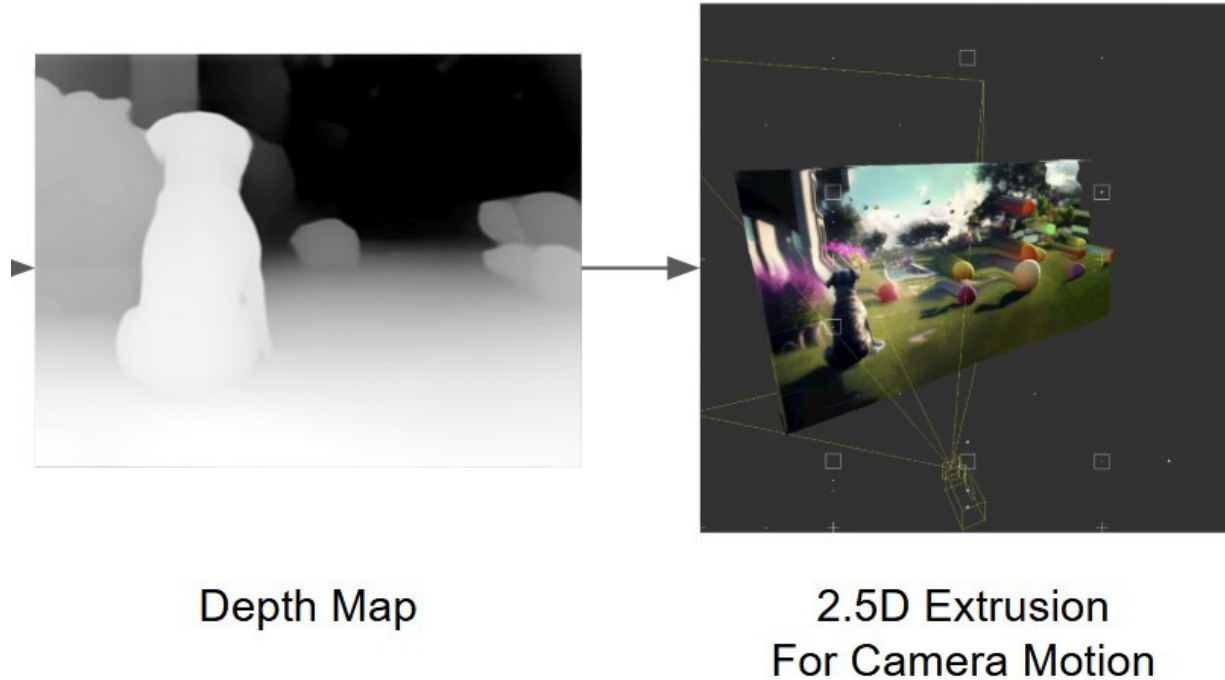
Zhang et al.
Generative AI for Film Creation
CVPR CVEU Workshop, 2025

A film combines several kinds of work



Images become part of a production workflow.

Depth can turn an image into a camera move



Near and far regions move differently as the camera shifts.

Film Hack evidence has a context

2023: 8 films

2024: 67 films

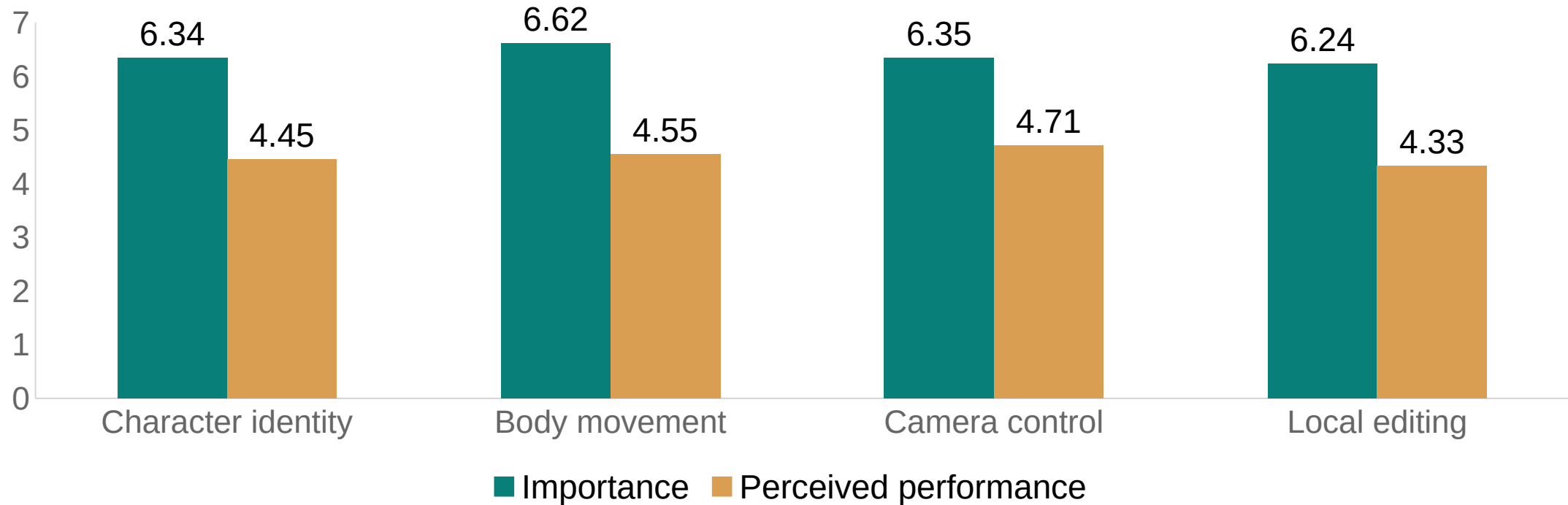
2025: 118 films

Creators combined tools

The 2025 sample averaged 3.14 video-generation tools per film.

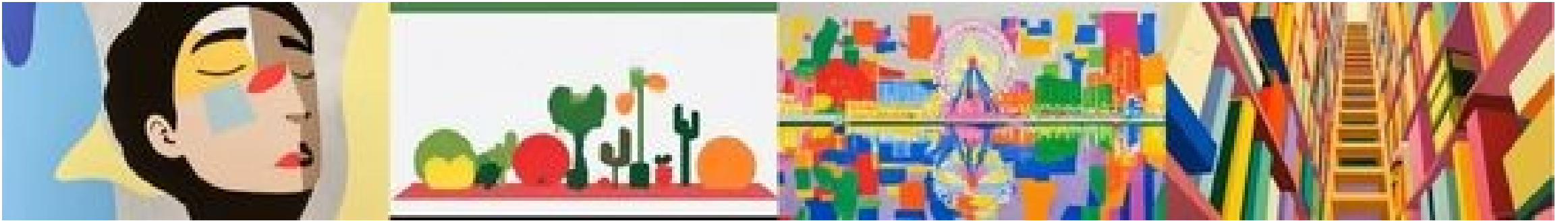
What filmmakers wanted

Importance exceeded reported performance in each selected task.



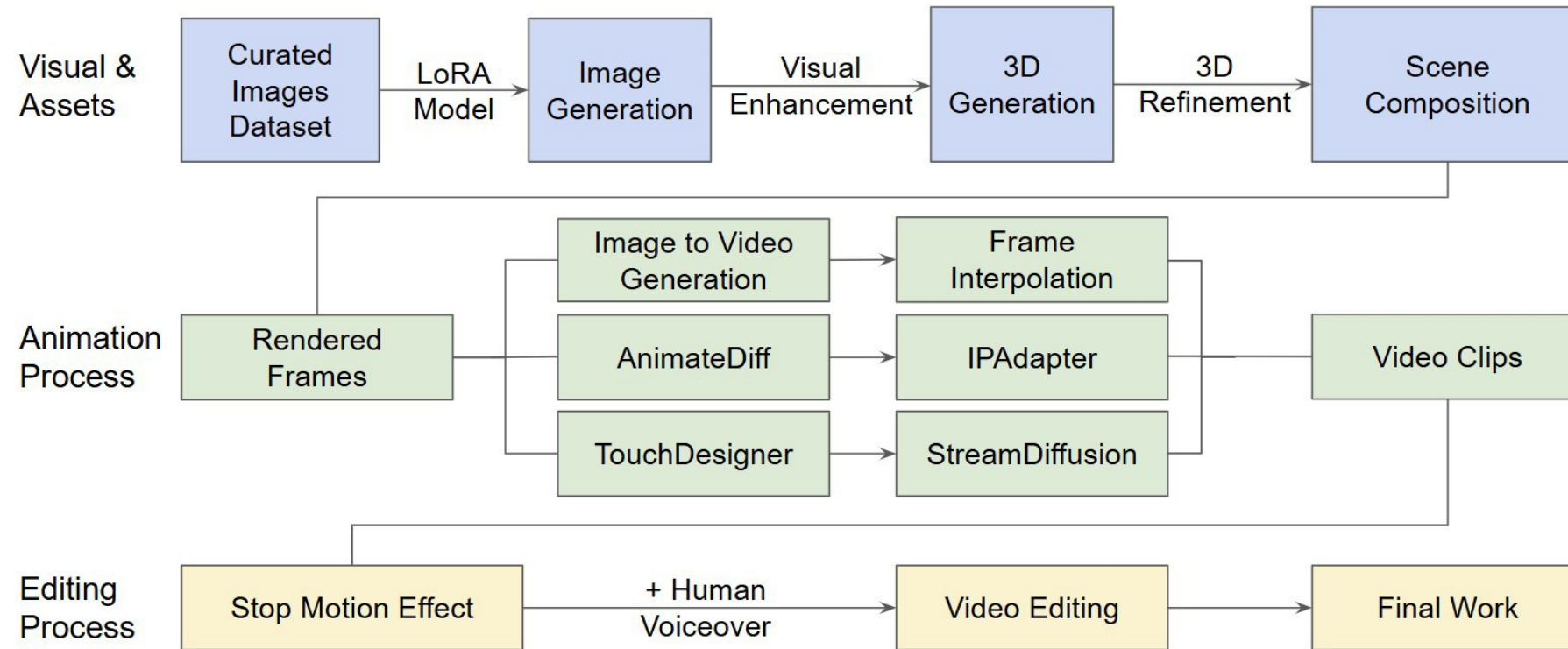
Artist ratings on a 0–7 scale. $n = 100$. Selected tasks from Table 5.

A visual style can begin with a drawing



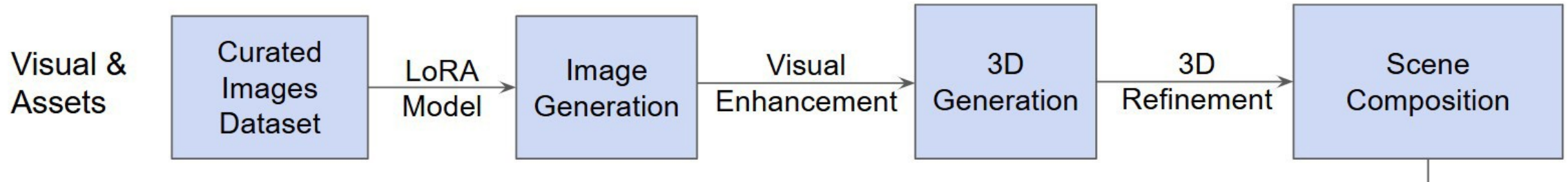
A Dream About to Awaken

Overthinking



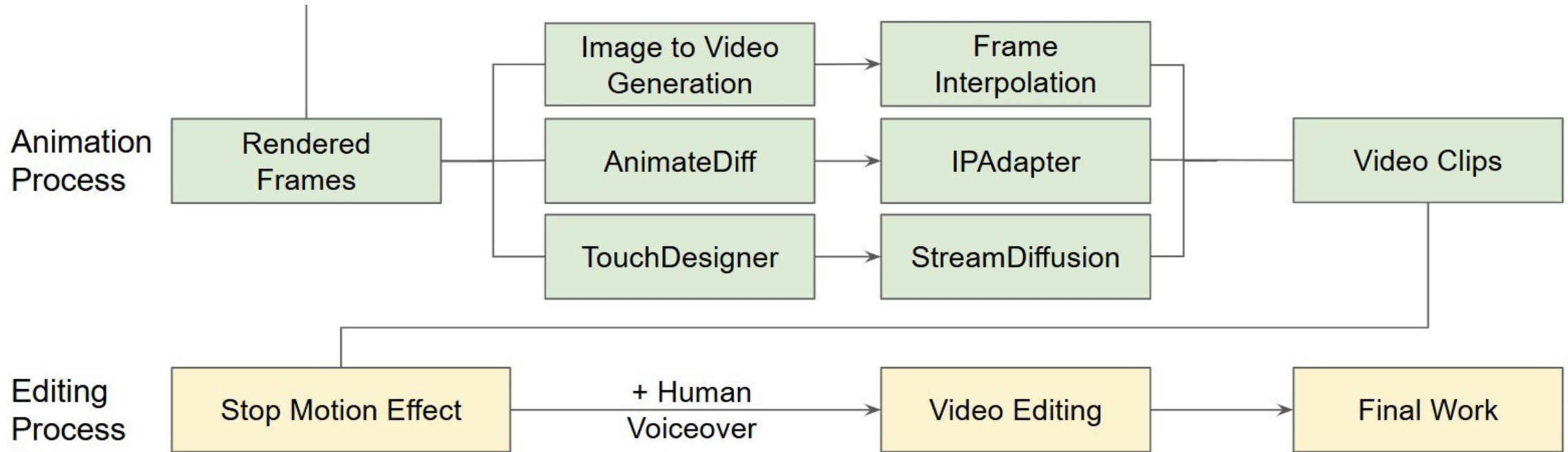
A deliberately nostalgic visual language

Style begins before animation



Overthinking builds assets before it builds movement.

Timing can carry a style



Overthinking also shapes its motion during editing.

Clown



Variation becomes part of the character's expression.

A source performance can survive a change of style



Compare the hand gesture across both rows.

Continuity can be a choice

Would a stable face or a changing face serve this scene?

For Pixi



A recurring character needs recognizable features.

A character reference is specific



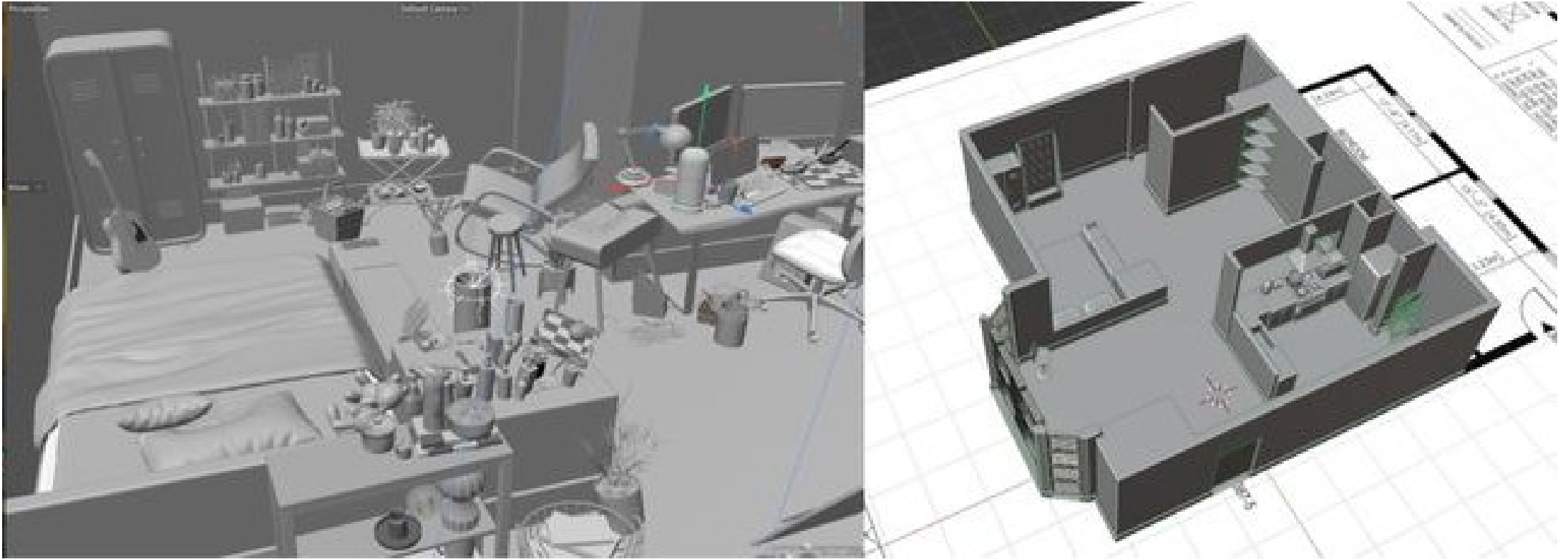
Which features make this character recognizable?

Fish Tank



Generated assets still enter a 3D production process.

A generated asset still needs production work



A scene needs usable geometry and consistent placement.

Volumography



A captured scene can support later camera decisions.

Fragmented space as an aesthetic



Gaps and distortions can suggest a memory.

A short film still needs decisions

What changes?

When does it change?

What should the audience feel?

A shot plan makes the next step concrete

Opening: establish the empty studio.

Change: reveal movement inside the painting.

Ending: show who returns.

Film-making checkpoint

A generated shot becomes useful when it serves the sequence.

04

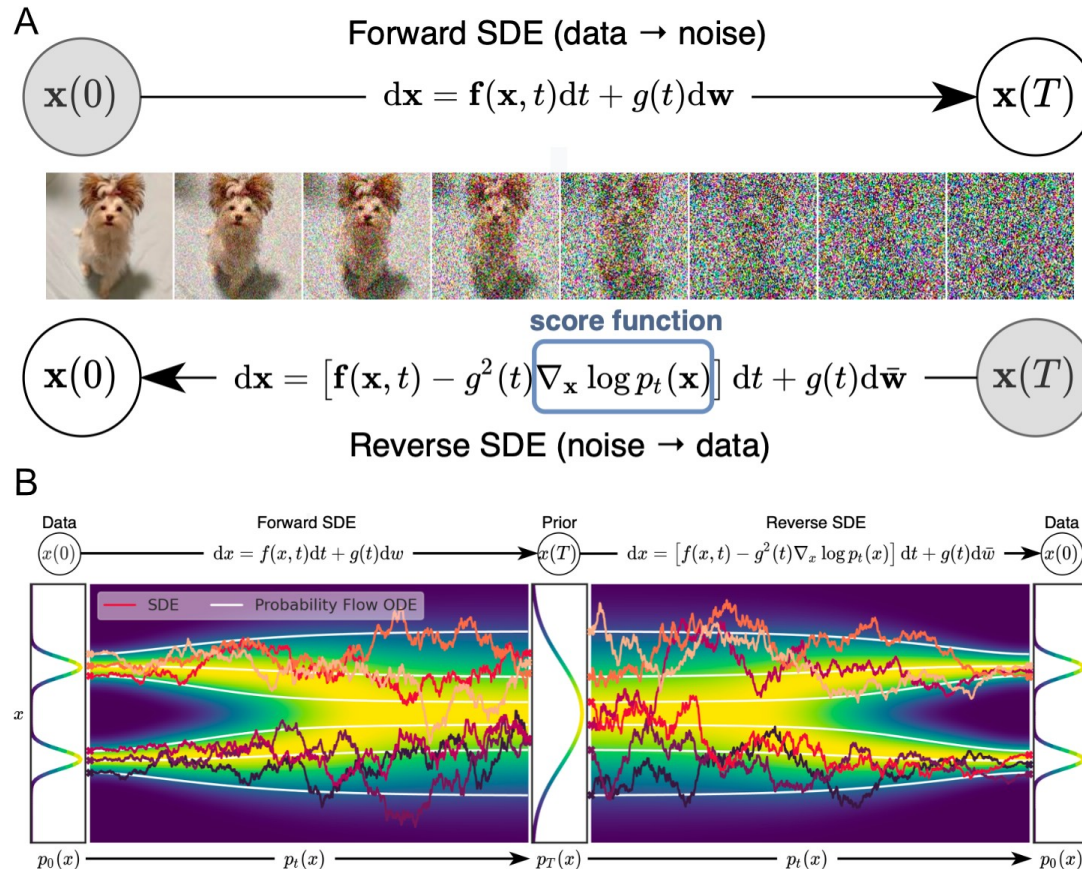
Visual computing

Po et al.

State of the Art on Diffusion Models

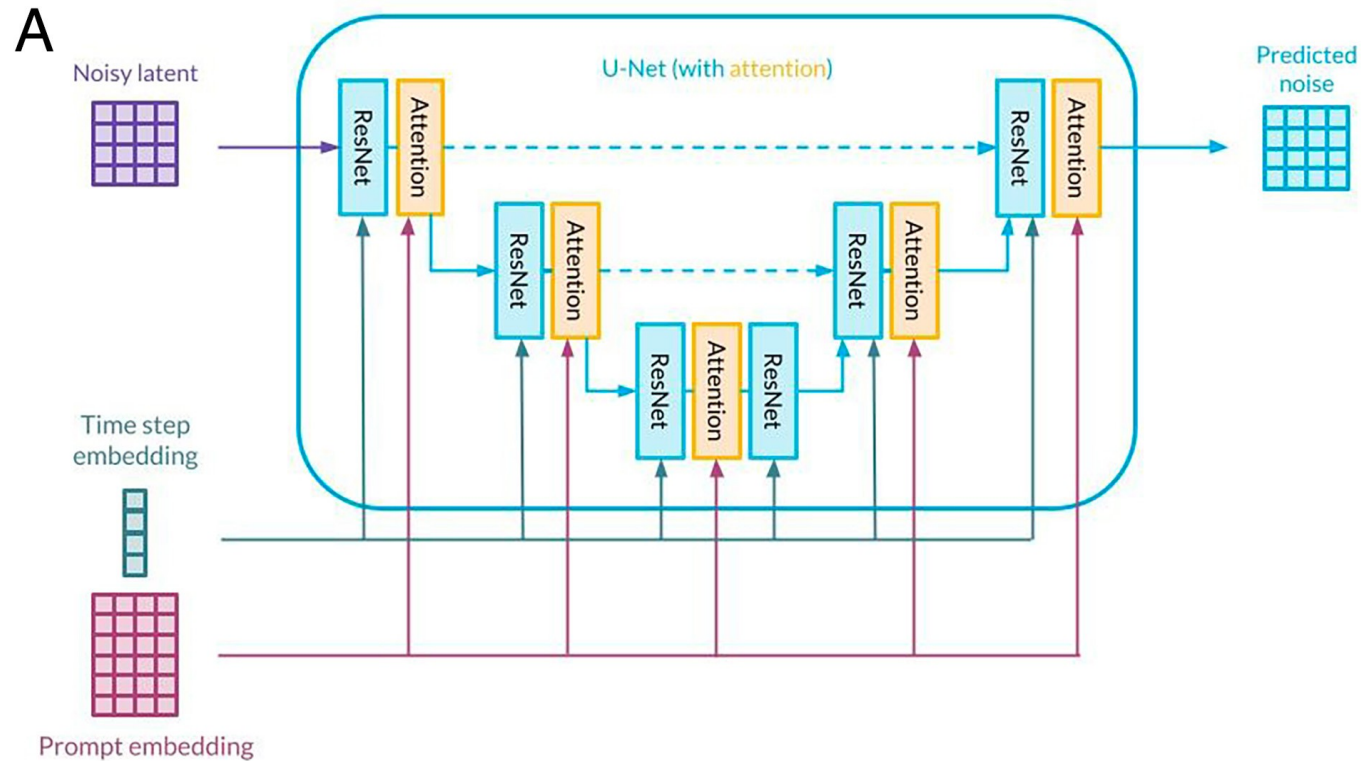
Computer Graphics Forum, 2024

The same process across visual media



A learned prior guides plausible visual outputs.

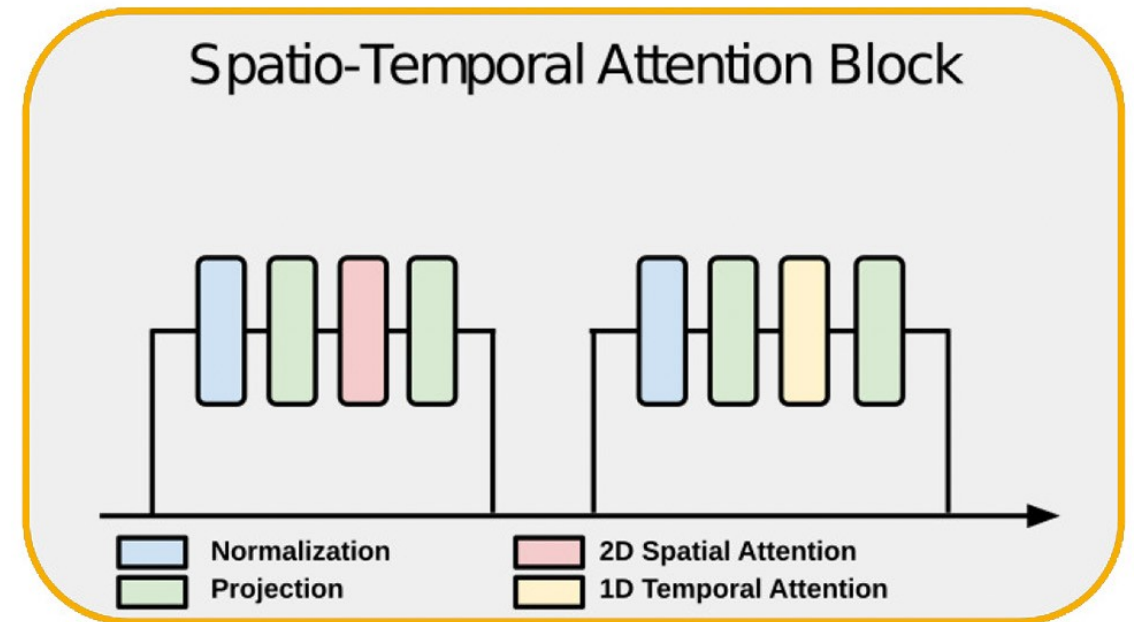
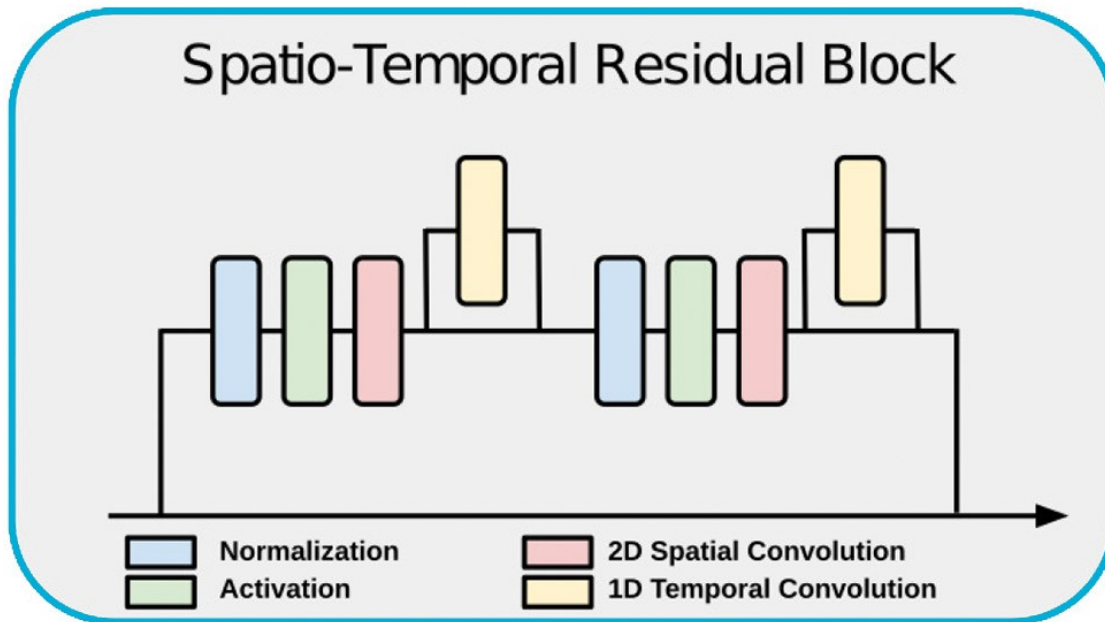
The denoiser uses the prompt at several scales



The network combines noisy content with conditioning.

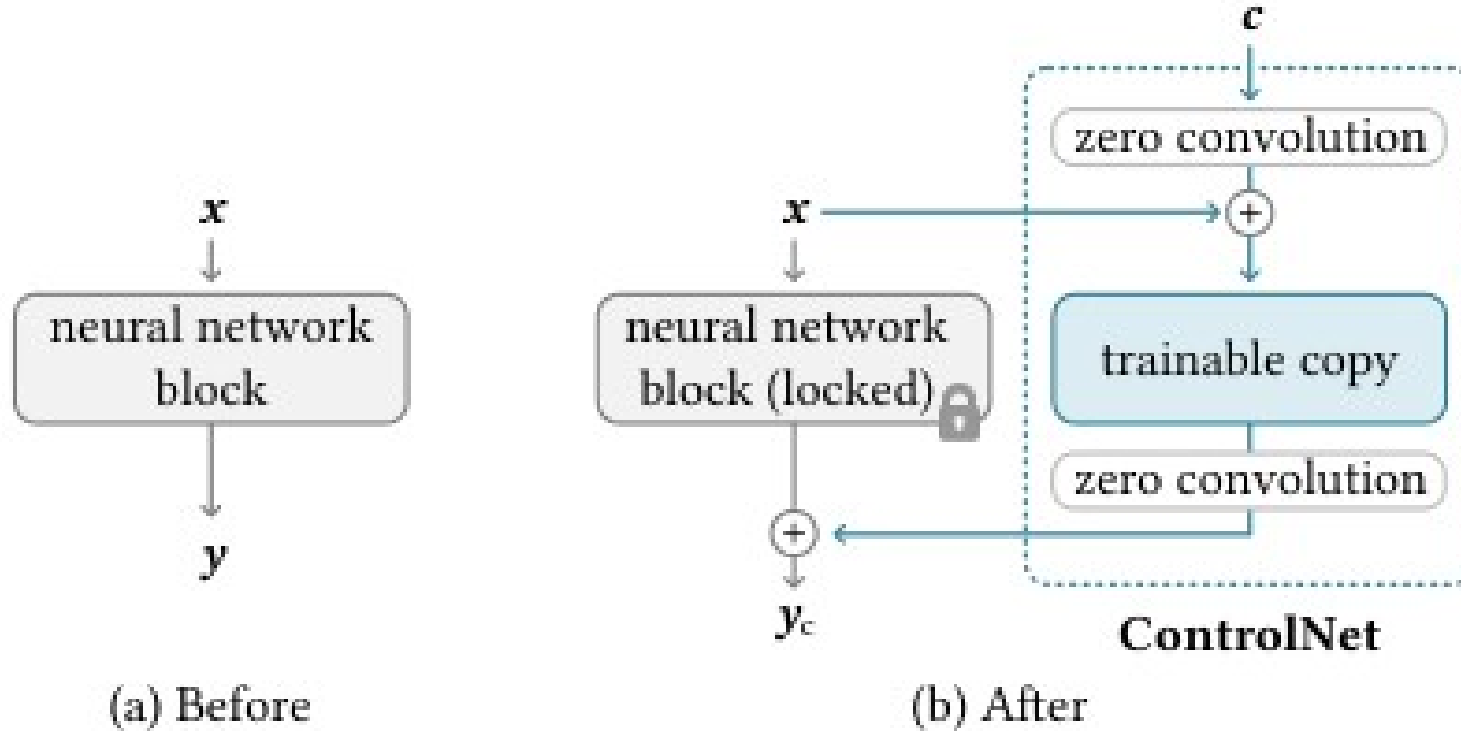
Temporal layers connect the frame features

B



An image model gains operations that work across time.

Conditioning adds a control signal



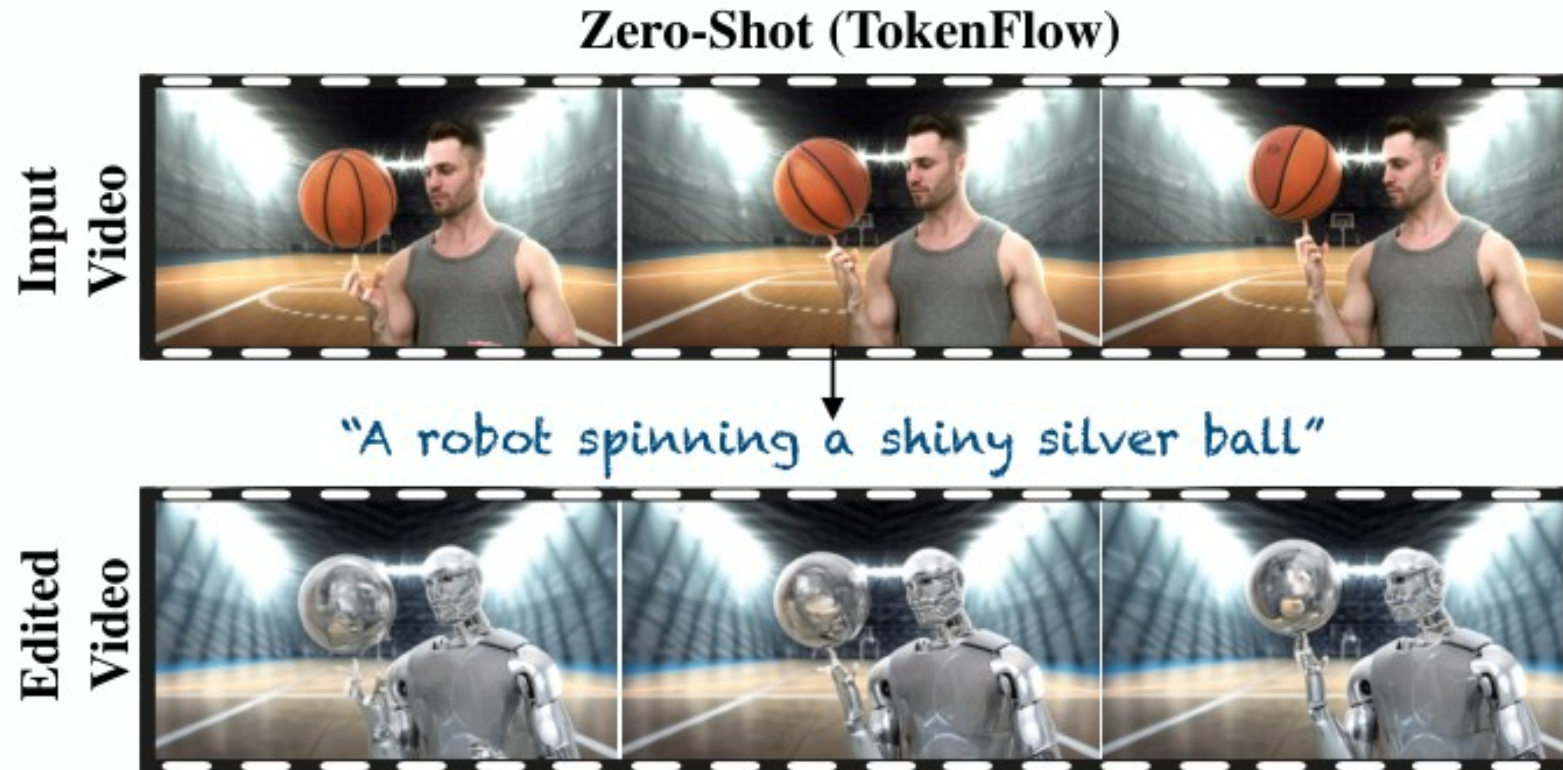
A sketch or depth map can guide structure.

Inversion and personalization

Inversion starts with an existing result.

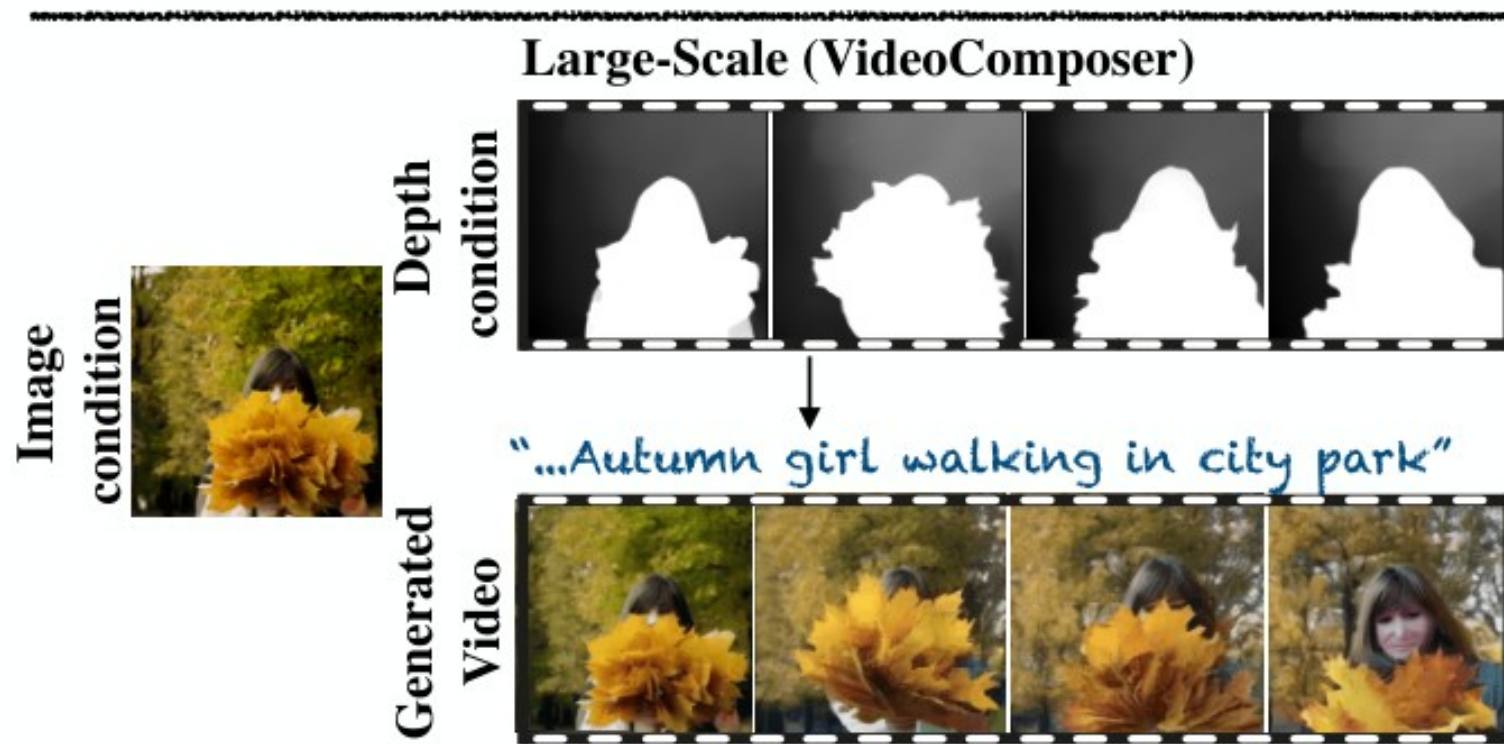
Personalization adapts to a subject or style.

Video editing can preserve movement



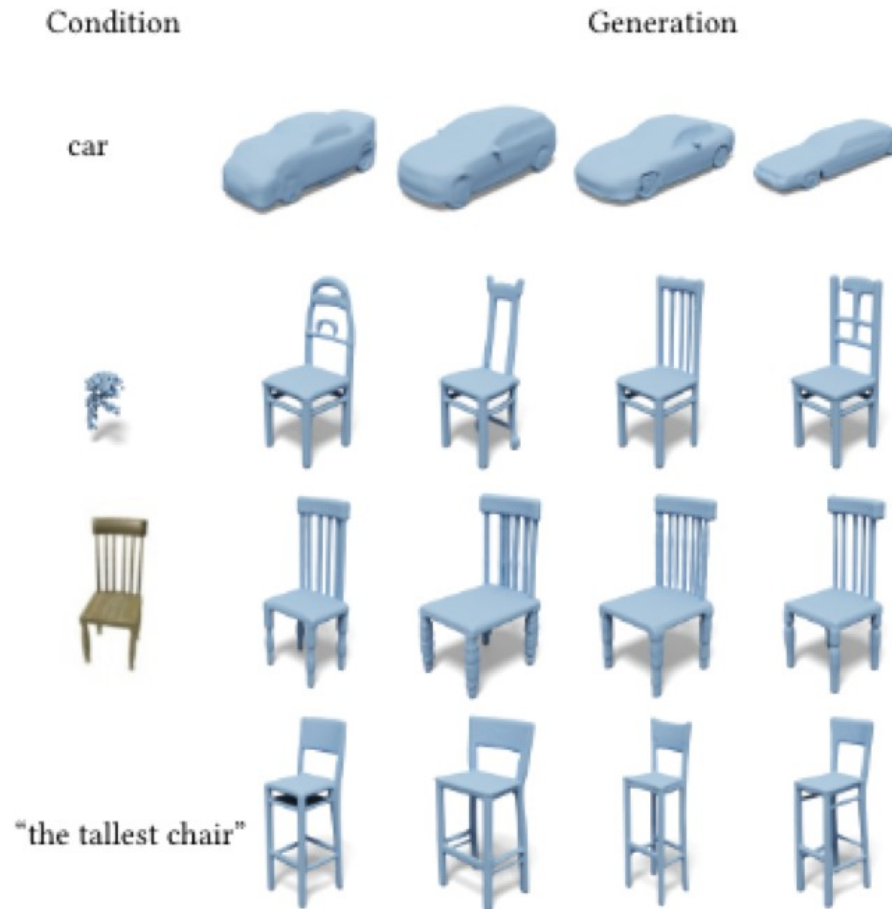
TokenFlow changes appearance across a source sequence.

Video generation can follow structure



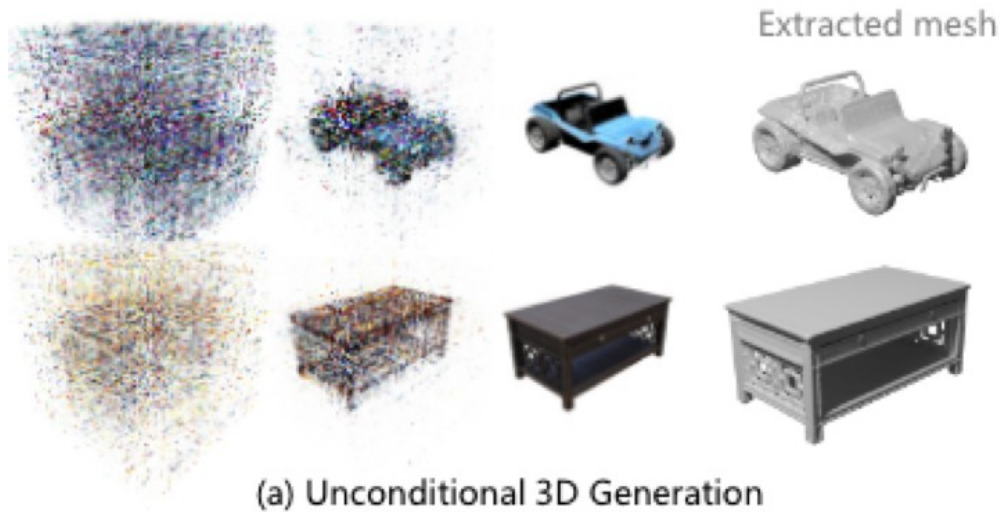
VideoComposer uses an image and depth information.

A 3D object has many views



Geometry must agree when the viewpoint changes.

Shape and appearance are different outputs



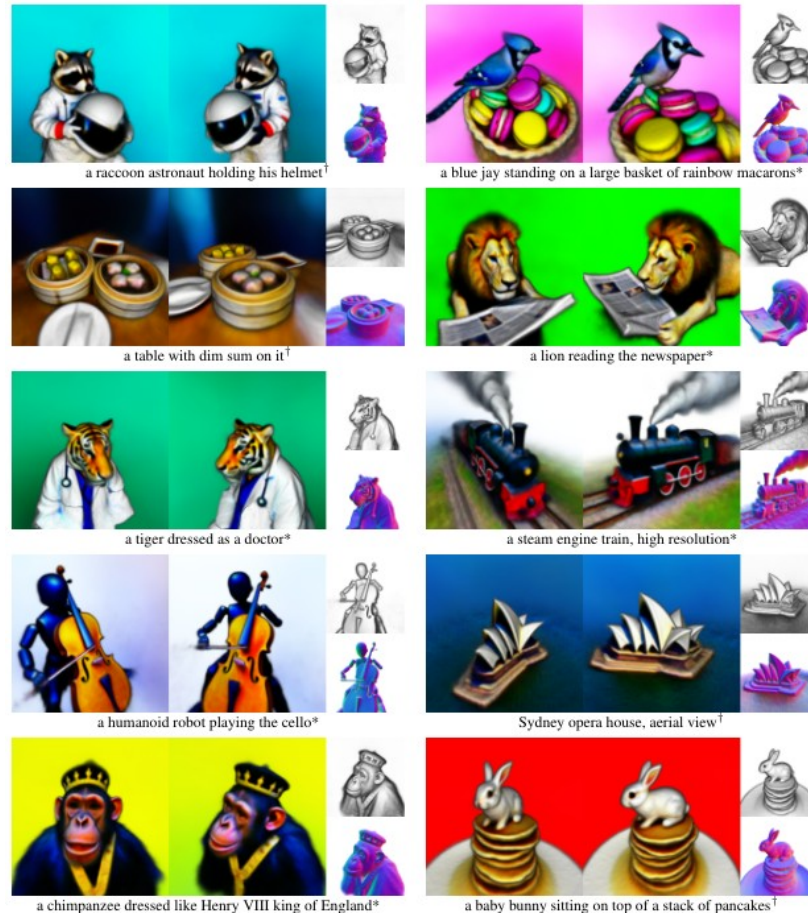
A mesh describes surfaces.
Appearance describes how they look.

A scene adds spatial relationships



A camera moves through a space with persistent structure.

Image models can guide 3D creation



DreamFusion uses image-based guidance to optimize a 3D result.

Editing a scene changes multiple views



InstructNeRF2NeRF edits a reconstructed scene.

The output format changes what we can edit

Pixels support direct image edits.

A scene representation supports new rendered views.

Motion can be generated as movement data



Audio can guide a dance before final rendering.

05

Creative industries

Anantrasirichai, Zhang, and Bull
Advances in Artificial Intelligence
Artificial Intelligence Review, 2026

Creation sits inside a larger workflow

Before generation: intention and references.

After generation: selection, editing, sound, and review.

One prompt can produce different images



(a) Prompt: A Japanese manga-style drawing of a boy wearing a school uniform and green shorts. He is jumping with his palms open in a field surrounded by flowers and butterflies.

(b)

Initial request

A realistic photo of a road scene with two cars. The car on the left is white. The car on the right is black.



ChatGPT4
+Dall E3 with
Manual
selection

Follow-up request

Change the
white car to a
red car

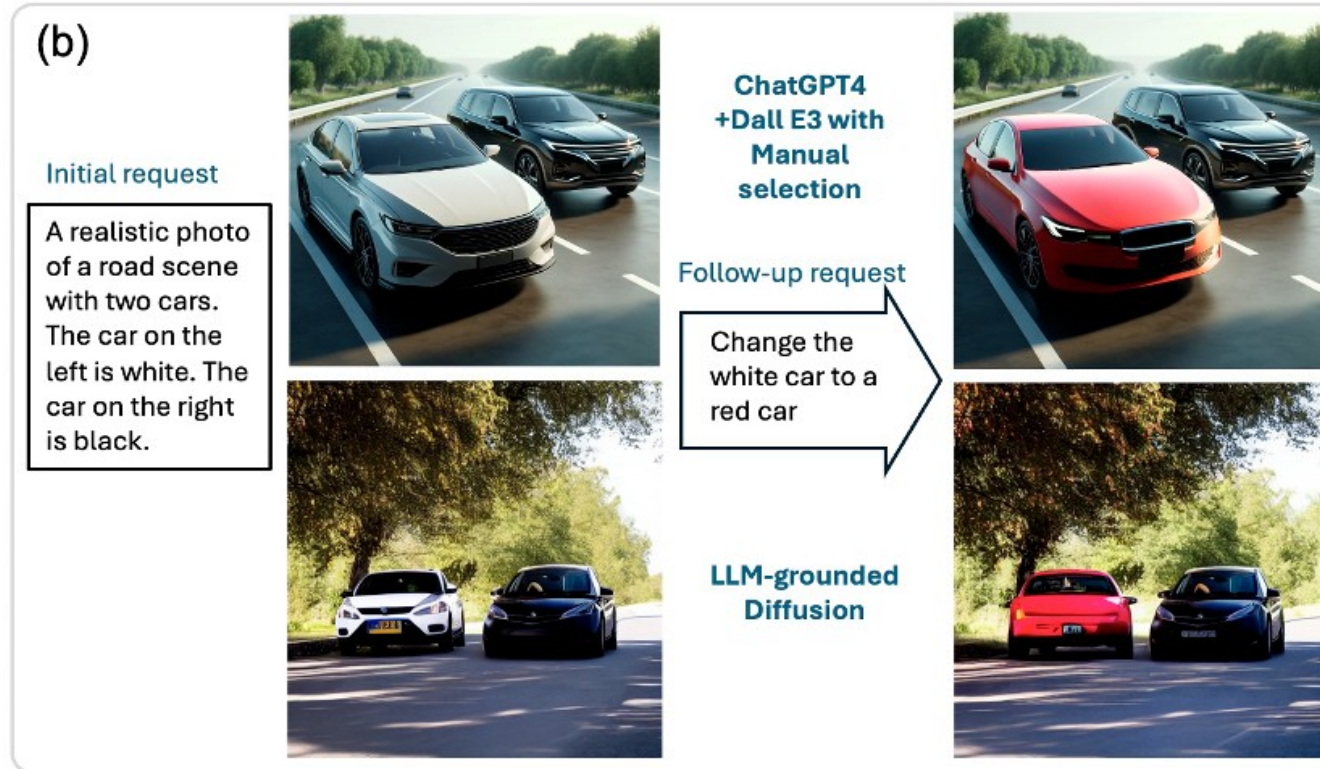


LLM-grounded
Diffusion



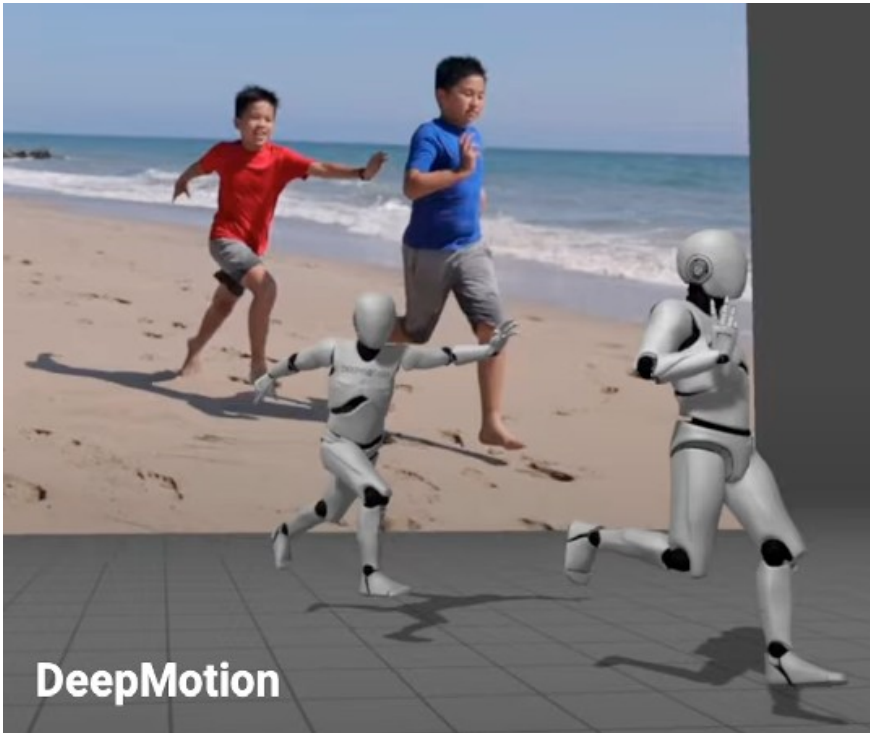
Models interpret composition and style differently.

A local instruction can affect the whole image



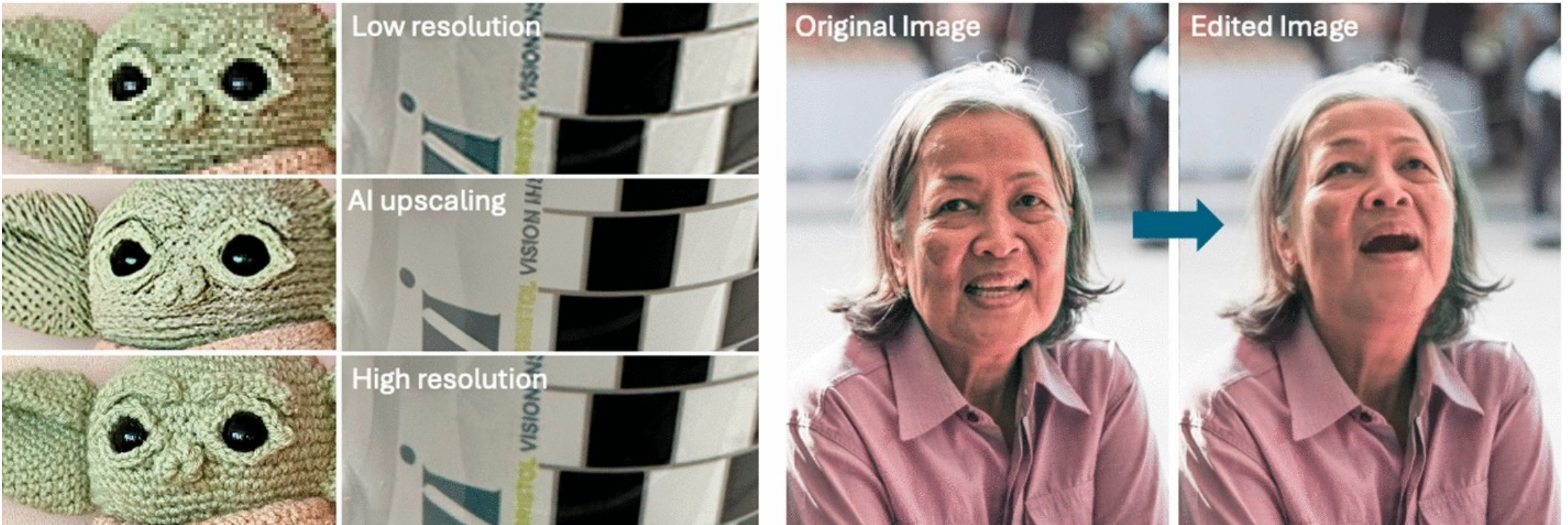
Compare the white car with its edited red version.

Performance can come from captured motion



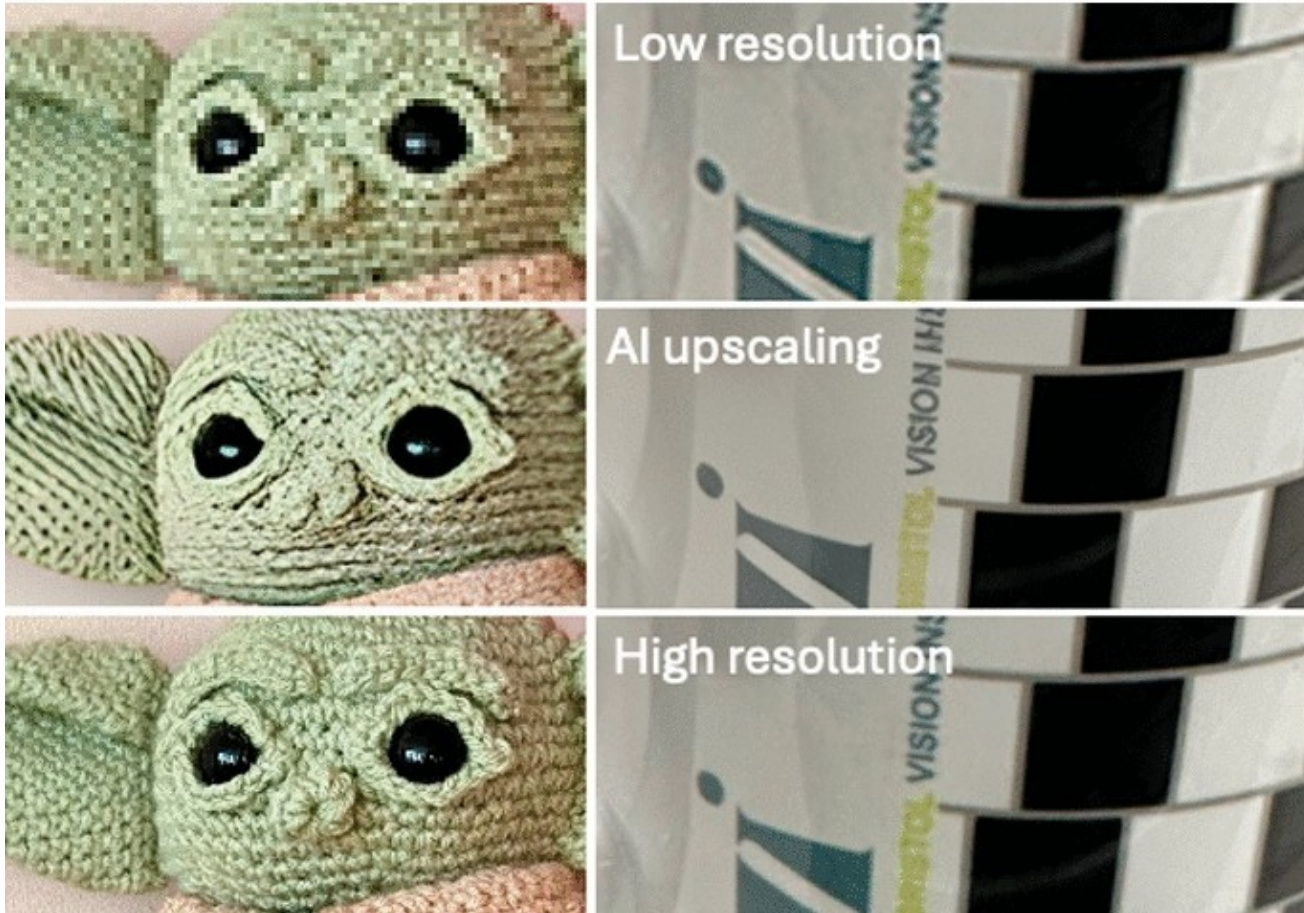
Body movement and facial animation use different inputs.

Post-production can change the result



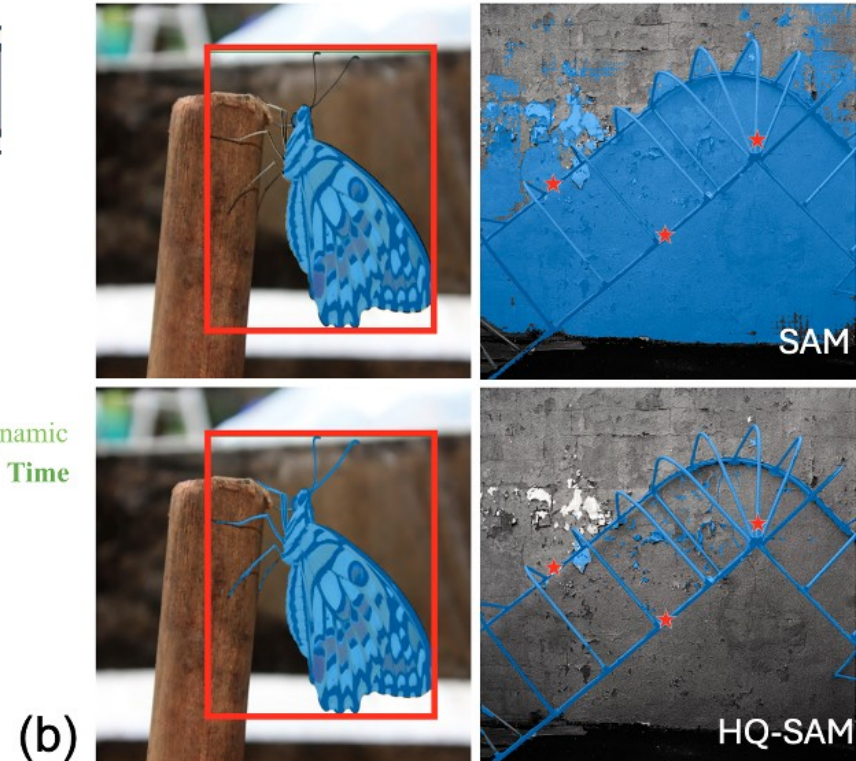
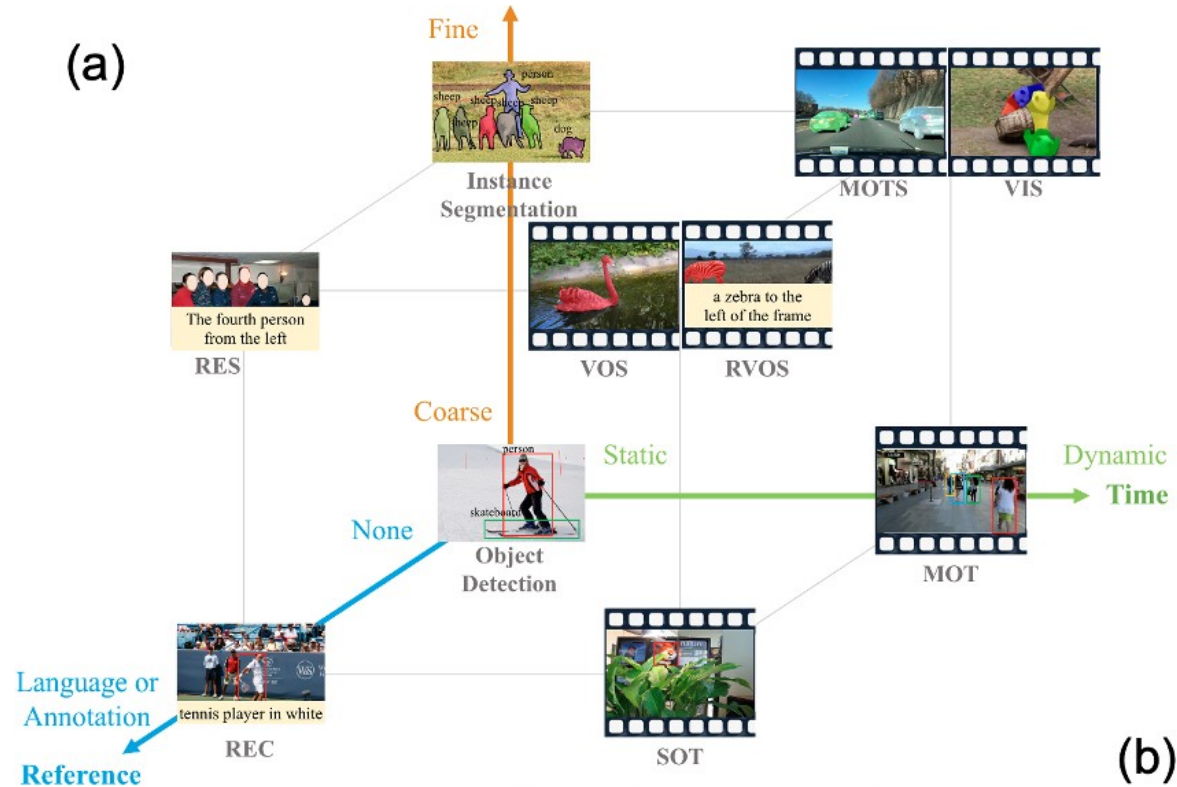
Enhancement and editing continue after generation.

Upscaling adds detail that needs inspection



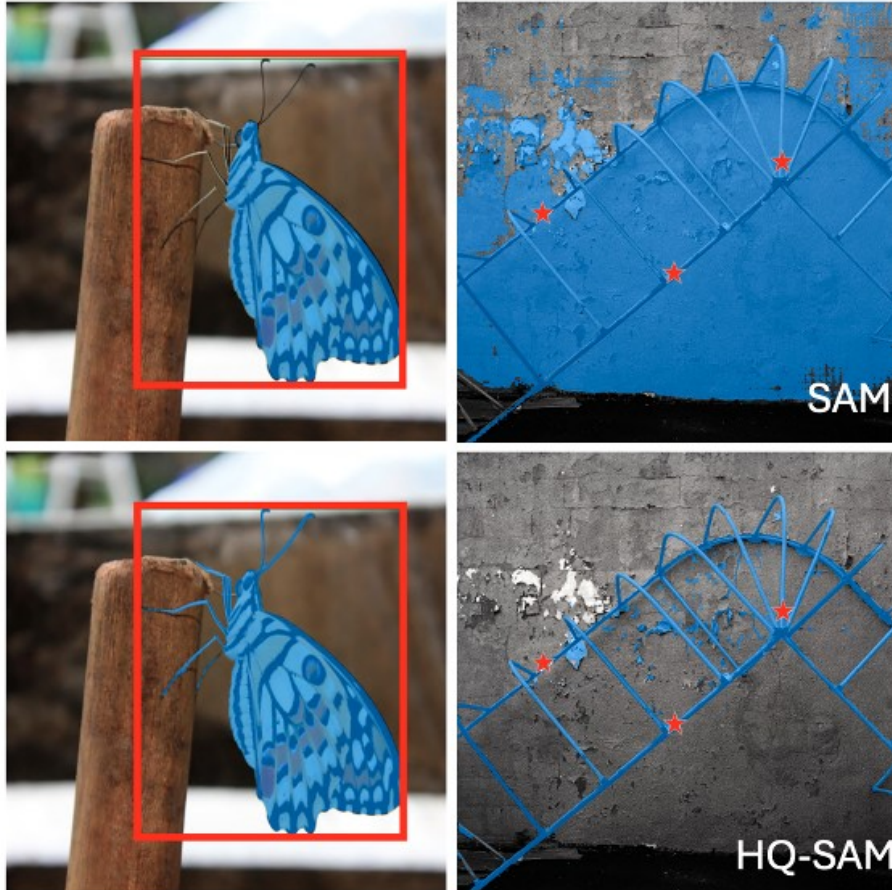
Compare the generated texture with the high-resolution reference.

Selection makes local editing possible



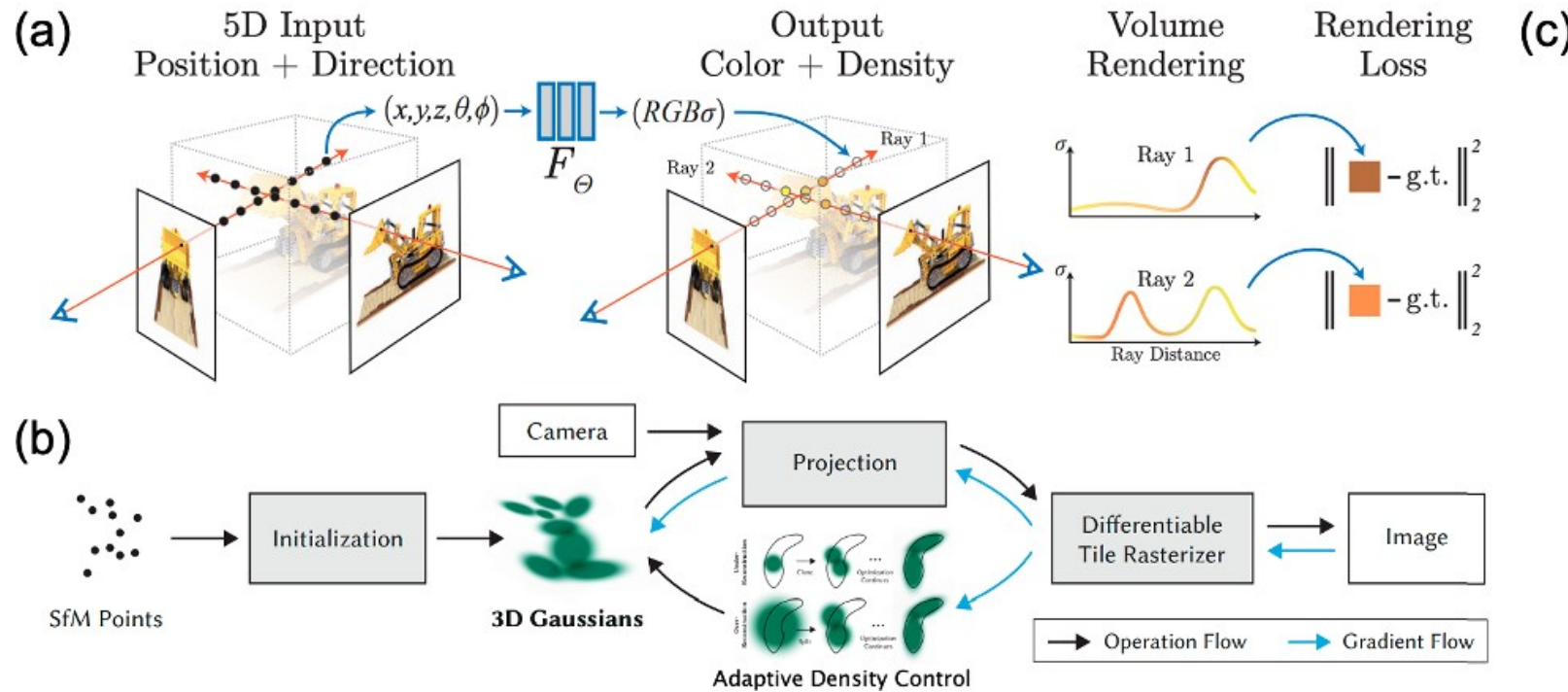
Segmentation identifies a region.
Tracking follows it through time.

Fine boundaries make selection difficult



Thin structures need careful masks.

Spatial capture supports new viewpoints



NeRFs and Gaussian splats represent scenes differently.

Quality has several meanings

A sharp image can still fail the scene.

A strange image can serve an artistic intention.

A process others can understand

Record references and major edits.

Explain the decisions behind the result.

Five readings to keep

- 01 Video models: how a sequence forms
- 02 Visual art: what the artist controls
- 03 Film: how shots work together
- 04 Visual computing: what transfers across media
- 05 Creative industries: how the work reaches an audience